Asymmetric Information and Capital Regulation in SME Lending: A Structural Model of Bank and Non-Bank Competition*

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Abstract

We study how competition between banks and non-banks in UK unsecured SME lending is shaped by capital requirements that apply only to banks, lenders' screening technologies, and their interaction. Using confidential loan-level data and new hand-collected data on screening channels, we document that banks screen smaller loans online and larger loans through relationship managers, while non-banks screen all loans online. To separate screening precision from borrower selection and from the cost of capital regulation, we develop a model of lending. Our estimates show that non-banks assess risk more precisely when both lenders screen online, but banks become more precise once they switch to human screening. In counterfactuals, tighter capital requirements mainly push the riskiest borrowers out of credit, but where banks screen imprecisely, they also lose some safe borrowers they misjudge as risky to non-banks. Moreover, if banks screened all loans online, more firms could borrow, but banks would screen less precisely, so their default rates would rise.

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1 Introduction

Non-bank lenders (non-deposit-taking institutions) have expanded rapidly in recent years and now compete directly with banks in many credit markets (Buchak et al., 2018; Cramer et al., 2025). One such market is lending to small businesses, which has long faced a finance gap. In the UK, the government puts this gap at £1.6 to £4.1 billion a year (HM Treasury, 2026). The gap has left room for non-banks to grow substantially (Gopal and Schnabl, 2022), so small businesses' access to credit now depends on the interplay between banks and non-banks.

Two salient features distinguish the two lender types. The first is regulatory: banks face risk-based capital requirements that do not apply to non-banks (Irani et al., 2021; Buchak et al., 2024). This regulatory asymmetry stems from their different funding. Banks rely largely on deposits, while non-banks raise funds through private credit markets (Dewatripont and Tirole, 1994). The second is technological: non-banks may adopt new technologies to a different extent (Fuster et al., 2019), which can translate into differences in screening ability. Crucially, the two are linked, because banks' capital requirements can depend on their own assessment of borrower risk.[1]

This paper asks how capital regulation, screening and their interaction shape interest rates and the allocation of credit and risk between banks and non-banks. We study the UK market for unsecured loans to small and medium-sized enterprises (SMEs), a key source of finance linked to firms' long-run growth (Aragoneses et al., 2026). This market suits our question for two reasons. First, screening is central. SME borrowers are informationally opaque (Petersen and Rajan, 1994; Liberti and Petersen, 2019), and their loans carry no collateral, which otherwise helps lenders deal with asymmetric information (Bester, 1985; Besanko and Thakor, 1987). Lenders' own risk assessment therefore matters even more. Second, the same lack of collateral makes these loans capital-intensive under risk-based requirements (Beaumont, Tang and Vansteenberghe, 2026), so regulation bites strongly in a setting where screening also matters.

We provide new evidence that differences in how lenders screen are closely linked to the

[1]This holds for banks using the internal ratings-based approach. The regulator allows these banks to use their own risk assessment models to calculate their capital requirements. All but a few of the banks that issue unsecured term loans to small businesses in our sample use this approach. Banks on the standardised approach, where the regulator sets risk weights, also appear in our data. We observe all their lending, and they originate very few unsecured term loans.

allocation of credit and risk between banks and non-banks. Using archived versions of lenders' websites, we hand-collect data on how lenders screen loan applications.[2] Banks screen loans online up to a bank-specific cap of either £25,000 or £50,000. Above the cap, they require borrowers to contact a relationship manager, so a person assesses the application. Non-banks screen all loans online. This switch coincides with a sharp change in market outcomes. Above the cap, the market share of banks falls relative to non-banks, and so does the default rate on their loans, even after accounting for observable firm characteristics. These patterns hold close to the cap as well, emerging sharply just above it rather than gradually with loan size.

Our descriptive analysis also shows that interest rates contain information about borrower risk beyond the borrower characteristics we observe. Among observably similar firms, those that pay higher rates default more often, for both banks and non-banks. Two explanations could account for this: lenders may observe risk that we do not and price it, or higher rates may themselves push firms into default. Our evidence favours the first (Appendix F).[3] Lenders thus price unobserved risk, and how closely their rates track default contains information on how well they screen: the more precisely a lender screens, the more its offered rates separate borrowers who will default from those who will repay. A second pattern links rates to capital regulation. Within banks, increases in capital buffers above the regulatory minimum are associated with lower rates, consistent with banks facing a cost from capital regulation. Because this cost scales with risk weights, it also makes bank rates rise more steeply with assessed risk. Rates on originated loans, and their correlation with default, therefore reflect three forces: screening precision, the cost of capital regulation, and how borrowers sort across lenders. We build a model that generates these descriptive patterns and lets us separate the three forces.

Our model centres on asymmetric information about borrower risk and builds on industrial organization models of lending (Crawford, Pavanini and Schivardi, 2018). On the demand side, borrowers differ in their default risk, which depends on observable characteristics and on a risk type that lenders do not observe, and need a loan whose size depends on the same characteristics and risk type. We take the screening structure as given: the size of the loan relative to the cap of the borrower's bank determines whether the bank screens it

[2]We observe the channel through which lenders accept applications and use it to infer how they screen; Section 2 describes this mapping.

[3]After the post-pandemic rise in interest rates, firms with floating-rate loans faced higher rates and their current account balances fell relative to firms with fixed-rate loans, a sign of a heavier repayment burden. Yet their default rates did not rise relative to those of fixed-rate borrowers.

online or through a relationship manager, and, consistent with our finding of no bunching below the cap that is correlated with risk, borrowers do not choose loan size to be screened by a particular bank channel. Each lender offers an individual rate based on its own assessment of the borrower's risk. Borrowers value both the rate and non-price features of each lender, which can differ between banks and non-banks and, for the bank, between its online and human channels. Applying through a relationship manager may involve non-price frictions, such as additional time and effort, and borrowers choose among the offers they receive or not to borrow.

On the supply side, the model has one bank and one non-bank, abstracting from heterogeneity among lenders of the same type. Our pricing and default regressions support this choice: replacing lender-type dummies with lender fixed effects barely changes the coefficients. For banks, borrowers also rarely shop around: more than 93% of bank loans are made by the bank where the borrower already holds its main current account. The two lenders compete on the rates they offer, and we let them differ in both screening precision and cost structure. Lenders screen through these rates rather than through approval: a lender that assesses a borrower as very risky offers a high rate, which plays the role of a rejection. Each lender learns about a borrower's unobserved risk type by Bayesian updating, combining the information in its loan size, which is correlated with risk, with a private screening signal. We estimate the precision of this signal separately for three screening technologies: the non-bank's online screening, and the bank's online and human screening. Banks additionally face a regulatory cost of capital that rises with each loan's risk weight. The interaction between capital regulation and screening works through this risk weight. Because it depends on the bank's own assessment of the borrower's risk, how precisely the bank screens determines its capital cost.

We estimate the model by indirect inference (Gourieroux, Monfort and Renault, 1993), matching pricing, default and loan size regressions, market shares and default rates in the data to their counterparts in simulated data. The model's demand side accounts for how borrowers select across lenders given the rates they are offered. Screening precision is then identified, in each segment, from how strongly each lender's rates co-move with the part of default that borrower characteristics do not explain, and from how often its borrowers default beyond what those characteristics predict. For banks, this co-movement also reflects the shadow cost of capital, which we identify separately from how bank rates vary with

observable risk. Finally, non-price frictions in the bank's human channel are identified from the fall in banks' market share above the cap that neither rates nor selection explain.

Our estimates show that the ranking of lenders' screening precision reverses at the cap, in a way that points to the type of information each screening technology relies on. Below the cap, where both lenders screen online, mainly on hard information, the non-bank screens more precisely than the bank. Above the cap, where the bank's assessment is led by a relationship manager who can also draw on soft information, the bank becomes the more precise lender. The estimated precisions imply correlations between a lender's assessment of a borrower's default probability and its true value of 0.16 for the bank's online screening, 0.32 for the non-bank, and 0.41 for the bank's human screening.

Turning to costs, we estimate that lending costs banks 4.85% a year and non-banks 8.60%. These costs combine origination costs, expected default losses and, for banks, the cost of capital regulation. Banks' advantage comes from lower origination costs, which likely reflect their cheaper deposit funding, and from lower expected losses. The same funding subjects banks to capital regulation, which accounts for 25% of their cost of lending and narrows their advantage without reversing it. This capital cost depends on how precisely the bank screens: if the bank screened as precisely as the non-bank, its capital cost below the cap would be about a quarter lower.

Our first counterfactual is motivated by bank branch closures and the shift of banking online. We move the bank's screening above the cap online, which removes the non-price friction of human screening but lowers the bank's screening precision for large loans. The share of firms that borrow above the cap rises by 25.2 percentage points, but the default rate on bank loans there rises by 0.97 percentage points. The credit expansion comes mostly from removing the friction, and the rise in default mostly from the lower precision. The link between regulation and screening appears again: because the capital charge depends on the bank's own risk assessment, the bank's risk-weighted capital cost above the cap rises by about 40%. Because the counterfactual treats the switch as costless and removes the friction of human screening entirely, its gains are an upper bound. Banks do appear to be moving in this direction: when a bank changes its cap during our sample, it raises it.

Our second counterfactual addresses a policy question: how do bank capital requirements affect small firms' access to credit? We raise banks' minimum capital requirement by 10

percentage points,[4] the benchmark increase in the literature (Hanson, Kashyap and Stein, 2011; Baker and Wurgler, 2015; Benetton, 2021). Banks' cost of lending rises by 83 basis points, and they pass about half of this on to borrowers. As a result, banks lose about 13% of their borrowers: about 10% stop borrowing and the other 3% switch to non-banks. Relative screening precision shapes who switches. Below the cap, where the bank screens online, the non-bank screens more precisely and attracts safer borrowers that the bank misjudged as risky and priced out. Above the cap, where the bank screens through relationship managers, the pattern reverses. The link between regulation and screening appears once more: how capital regulation reallocates risk depends on screening precision.

Related Literature This paper contributes to several strands of literature. First, we contribute to the literature on competition between banks and non-banks. One part of this literature studies how bank regulation shapes this competition (Irani et al., 2021; Chernenko, Erel and Prilmeier, 2022; Buchak et al., 2024), another how non-banks' technology does (Fuster et al., 2019). Buchak et al. (2018) quantify both forces in mortgages, and Cramer et al. (2025) show that technology matters more in unsecured lending and regulation in secured lending.[5] In these papers, technology mostly means convenience, faster processing or easier access, rather than how precisely lenders assess borrower risk. Di Maggio and Yao (2021) use a notion closer to ours, comparing the screening technology of fintech and traditional lenders in US personal loans. We build on this by studying screening technology and capital regulation together, accounting for the link between them. We estimate each lender's screening precision, including online and human screening within the same bank, and show how it shapes competition between banks and non-banks.

Second, we contribute to the literature on screening in credit markets, which builds on work on asymmetric information in lending (Stiglitz and Weiss, 1981; Crawford, Pavanini and Schivardi, 2018).[6] Recent work studies screening in lending markets where lenders

[4]This raises the requirement from about 13% to 23% of risk-weighted assets. For comparison, removing the SME supporting factor, as discussed in the UK's implementation of Basel 3.1, would on its own be equivalent to about 40% of this shock (Prudential Regulation Authority, 2025).

[5]See also Gopal and Schnabl (2022), Tang (2019), Lee, Lee and Paluszynski (2024), Aldasoro, Doerr and Zhou (2025), Fleckenstein et al. (2025), Lyonnet and Chrétien (2025) and Benetton, Mayordomo and Paravisini (2025).

[6]See also Petersen and Rajan (1994), Sufi (2007), Ivashina (2009), Einav, Jenkins and Levin (2012), Iyer et al. (2016), Darmouni (2020) and Bosshardt, Kakhbod and Kermani (2025). Stillerman (2024) studies how government guarantees, which insure lenders against default, affect their incentives to screen SME borrowers.

post prices: through menus of contracts that borrowers self-select into (Polo, Taburet and Vo, 2025), or through risk-based credit limits, with borrowers choosing among posted rates (Matcham, 2025). In our setting, lenders instead set each borrower's rate individually, based on their own assessment of risk. Outside lending, Cosconati et al. (2025) estimate differences in the precision of insurers' risk ratings when prices are set individually. We estimate screening precision in a lending market, comparing it across lenders and, within the same bank, across online and human screening. Our finding that precision depends on the screening technology relates to work on hard and soft information in lending (Stein, 2002; Berger and Udell, 2006; Agarwal and Hauswald, 2010; Hertzberg, Liberti and Paravisini, 2010; Liberti and Petersen, 2019) and on how loan officers and automated models assess risk (Paravisini and Schoar, 2013; Berg, Puri and Rocholl, 2020; Berg et al., 2020; Fuster et al., 2022).

Third, we contribute to the literature on how bank capital requirements affect credit markets.[7] The closest work estimates an equilibrium model of the UK mortgage market, where a loan's risk weight depends mainly on observable characteristics such as loan-to-value (Benetton, 2021). Lenders in that market also price mainly on these observable characteristics (Robles-Garcia, 2019; Benetton, Gavazza and Surico, 2025). In unsecured SME lending, both the rate and the risk weight instead depend on the bank's own assessment of each opaque borrower. The cost of regulation therefore is determined by how well the bank screens. We estimate this cost and show that the effect of a higher capital requirement on credit and risk allocation depends on screening precision. We assume that risk weights reflect banks' true assessments, abstracting from evidence that banks can use internal models to lower capital charges (Behn, Haselmann and Vig, 2022).

Finally, we contribute to the growing literature that applies industrial organization methods to financial markets.[8] Much of this work studies credit markets where rates are posted on observable characteristics (Benetton, 2021; Benetton, Gavazza and Surico, 2025; Nelson, 2018; Galenianos and Gavazza, 2022; Fisher et al., 2024; Benetton and Buchak, 2024) or negotiated (Allen, Clark and Houde, 2014, 2019). The closest antecedent is Crawford, Pavanini and Schivardi (2018), who study adverse selection and market power in Italian SME lending and recover borrowers' unobserved competing offers in reduced form, using borrower and

[7] See, for example, Peek and Rosengren (1995), Repullo (2004), Gropp et al. (2019), De Jonghe, Dewachter and Ongena (2020), Benetton et al. (2021), de Ramon, Francis and Harris (2022) and Plosser and Santos (2024).

[8] Beyond lending, this work studies pensions (Hastings, Hortaçsu and Syverson, 2017) and insurance (Koijen and Yogo, 2016).

bank-region-year fixed effects. We instead model the screening that generates these offers, with a precision that can differ across lenders and screening technologies. This matters in our setting because a bank's capital charge depends on its assessment of each borrower's risk, which fixed effects for offers do not reveal.

The paper proceeds as follows. Section 2 describes the institutional setting and data, followed by the motivating empirical facts in Section 3. Section 4 develops the model. Section 5 details the estimation strategy and presents the estimation results. Section 6 outlines our counterfactual analysis. Section 7 concludes.

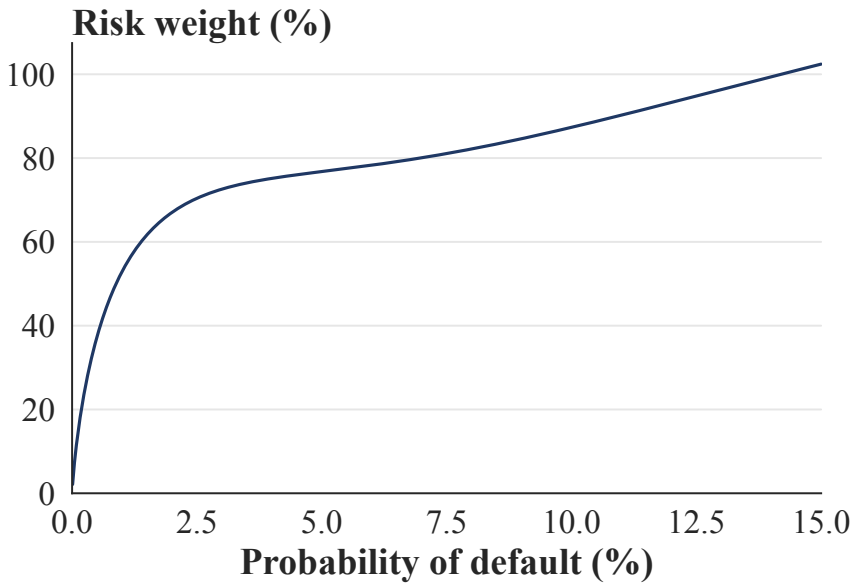
2 Institutional Setting and Data

2.1 SME Lending and Risk-Based Capital Requirements

Access to finance for small and medium-sized enterprises is a long-standing focus of UK policy (British Business Bank, 2023), and the effect of bank capital regulation on SME lending is a recurring part of that debate. In the UK's implementation of Basel 3.1, for example, concern about SME lending shaped the debate over whether to keep a discount on the capital required against SME loans (Prudential Regulation Authority, 2025). Capital regulation aims to make banks able to absorb losses and so protect financial stability. Banks must fund part of their lending with their own equity capital, not only with deposits or debt. Under risk-based capital regulation, the required capital is a minimum ratio of the bank's risk-weighted assets, the sum of its loans weighted by their risk (de Ramon, Francis and Harris, 2022). Each loan therefore adds to the bank's capital requirement in proportion to its risk weight.

Regulation allows two ways to calculate risk weights. Under the standardised approach, used mostly by smaller banks, each type of exposure gets a fixed risk weight. Unsecured loans to SMEs, for example, typically get 75%. Under the internal ratings-based (IRB) approach, banks estimate each loan's probability of default with their own regulator-approved models, and a regulatory formula turns this estimate into a risk weight (Appendix A). As Figure 1 shows, the risk weight rises with the estimated probability of default. The capital a loan requires therefore rises with the bank's own assessment of the borrower's risk. Almost all the banks that originate unsecured SME loans in our sample use the IRB approach.

FIGURE 1: Default Risk and Regulatory Risk Weights



Note: Regulatory risk weight of an unsecured SME retail loan under the IRB approach, as a function of the bank's estimated probability of default (PD). The risk weight follows the formula for other retail exposures in chapter CRE31 of the Basel Framework, with the SME supporting factor applied (Appendix A). The figure uses the bank's loss given default in the model, 68%.

Throughout the paper, "bank" refers to an IRB bank. Two facts suggest that IRB banks account for most bank lending in our market. First, in the Bank of England's regulatory data (Section 2.2), IRB banks account for about two-thirds of all SME lending by UK banks, including secured lending. Second, a bank in our loan data reports every product its SME customers hold, including current accounts, loans, and credit cards. In these data, banks that use the standardised approach are active in other products, and the loans they make are mainly secured. They originate very few unsecured term loans.[9] The two-thirds share is therefore likely a lower bound on IRB banks' share of unsecured SME term lending.

Regulation and screening are linked through risk weights in this market. The risk weight depends on the bank's own estimate of the probability of default. The capital a bank holds against a loan, and hence its cost of capital regulation, thus depends on its risk assessment of the borrower. In contrast, non-banks are not directly subject to these requirements, as they fund their lending through investors rather than deposits.[10] This regulatory asymmetry, and

[9]One reason may be that a flat risk weight gives a bank using the standardised approach a higher capital charge than an IRB bank on safe borrowers, which makes it harder to compete for them.

[10]Non-banks could still be indirectly affected by bank capital regulation if banks were major providers of their funding, since banks' capital charges on that funding would raise non-banks' cost of lending. Based

the link between capital regulation and screening, are the focus of this paper.

2.2 Data

We combine four data sources: loan data on UK SMEs, firms' annual accounts and credit scores, banks' regulatory data, and a new dataset on how lenders accept loan applications.

Loan data. Our main data source is a monthly panel of SMEs' financial accounts, including current accounts, secured and unsecured loans, and credit cards, reported to the Bank of England by Experian, a credit reference agency, under the Commercial Credit Data Sharing scheme.[11] The data cover SMEs with annual turnover below £25 million and their accounts with major banks and non-banks. We focus on unsecured term loans. These loans matter most for the smallest firms, which typically lack collateral and are the most opaque. Among all registered UK companies with debt, about four in ten micro firms hold at least one unsecured term loan, compared with three in ten small firms and one in five medium-sized firms.[12] For each loan, we observe the lender, the origination date, the maturity, the monthly outstanding balance and scheduled payment, and whether the loan defaults. We define a loan as defaulted if it becomes more than 90 days delinquent within one year of origination. We recover each loan's interest rate from the evolution of its outstanding balance and scheduled payments. We also use the current account data to identify each borrower's main bank, the bank that holds the largest share of its current account activity.

Firm characteristics. We link each borrower, through its company registration number, to the annual accounts it files with Companies House, accessed through Bureau van Dijk, which provide firm age, total assets, location and industry. We also obtain each firm's credit score from CRIF through Bureau van Dijk. Scores range from 0 to 100, with higher values indicating a lower probability of insolvency.

on the public information we reviewed, however, the investors funding the non-bank lenders in our data are mostly asset managers, and banks do not appear to be major investors.

[11] The scheme was introduced by the Small and Medium Sized Business (Credit Information) Regulations 2015, under the Small Business, Enterprise and Employment Act 2015.

[12] Firm size categories follow the UK Companies Act thresholds in force over our sample period. A firm is micro, small or medium-sized if it meets at least two of the following: turnover of at most £632,000, £10.2 million or £36 million; a balance sheet of at most £316,000, £5.1 million or £18 million; and at most 10, 50 or 250 employees, respectively.

TABLE 1: Summary Statistics

	Mean	Median	P25	P75
Loan characteristics				
Loan amount (£000s)	83.2	50.0	21.4	106.5
Interest rate (%)	12.03	11.00	8.98	15.40
Maturity (months)	58.0	60.0	36.0	72.0
Firm characteristics				
Total assets (£000s)	602.8	175.0	60.0	542.0
Firm age (years)	11.0	9.0	5.0	14.0
Number of loans		58,291		

Note: Summary statistics for the estimation sample. P25 and P75 denote the 25th and 75th percentiles.

Bank regulatory data. For banks, we use the Bank of England's Historical Banking Regulatory Database, which contains confidential regulatory data including each bank's minimum capital requirement and its actual capital ratio (de Ramon, Francis and Milonas, 2017). We classify a lender as a bank if it appears in this database, and as a non-bank otherwise.

Screening channels. We construct a new dataset on how lenders accept loan applications. We scrape the websites of the bank and non-bank lenders in our sample over time, using archived versions of their pages from the Internet Archive's Wayback Machine. At the lender-quarter level, we record whether applications are submitted online or through a relationship manager, and the largest loan accepted online, which we call the online application cap. Section 2.3 describes the pattern these data reveal.

Sample. Our sample covers loans originated between the second quarter of 2021 and the fourth quarter of 2024, excluding loans under government-guaranteed schemes.[13] The final sample contains 58,291 loans. Table 1 reports summary statistics. The median loan is £50,000, and the distribution of loan sizes is right-skewed, with a mean of £83,237. Borrowers' total assets range from £60,000 at the 25th percentile to £542,000 at the 75th. The median interest rate is 11.0%, with an interquartile range from 9.0% to 15.4%.

[13]The COVID-era government-guaranteed loan schemes stopped accepting new applications at the end of March 2021, before the start of our sample. Our data identify loans originated under these schemes separately, and we exclude them.

2.3 Screening Channels in UK SME Lending

Our scraped data reveal a feature of UK SME lending that, to our knowledge, is new: banks screen loans through different channels depending on their size. Every bank in our sample accepts online applications up to a loan size cap of either £25,000 or £50,000, and requires borrowers above the cap to apply through a relationship manager. Non-banks accept online applications at every loan size.

When applications are made online, we call the screening *online screening*: the lender relies mainly on automated tools and hard information, although a person may still play a role. When a relationship manager handles the application, we call it *human screening*: the relationship manager is mainly responsible for the assessment and can also draw on soft information, although they may use automated tools.

The switch from online to human screening at the bank's cap coincides with markedly different market outcomes on either side of the cap, which we document in Section 3.

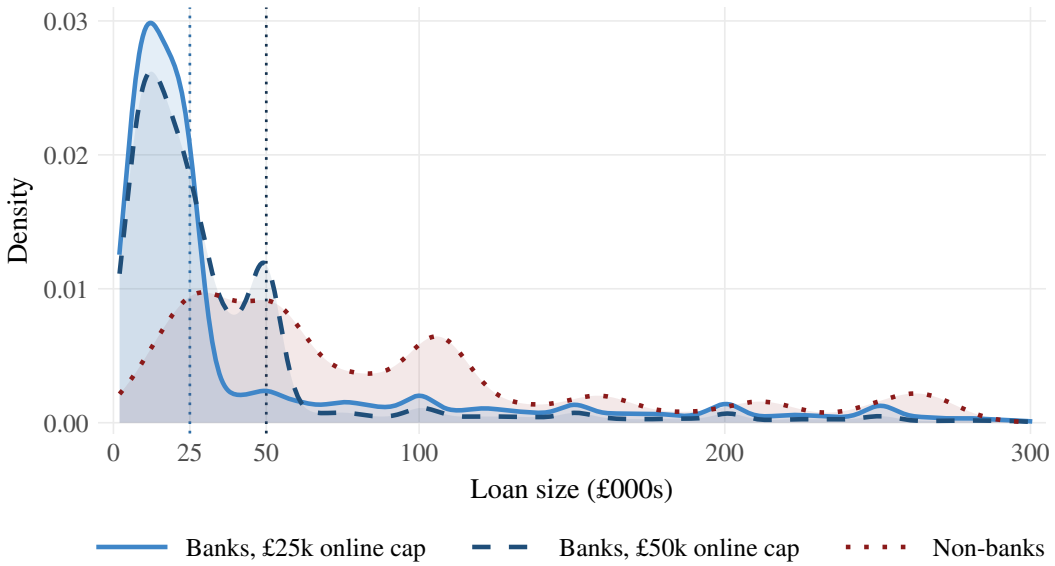
3 Motivating Facts

The goal of this section is to document the main patterns in how banks and non-banks screen, price and share risk in UK unsecured SME lending. We documents four facts. First, building on the screening pattern documented in Section 2, banks' market share and default rates fall sharply relative to non-banks above the online application cap, where banks switch to human screening. This gap in default rates persists even conditional on observable borrower characteristics. Second, interest rates contain information about default risk that our observable firm characteristics do not capture. Third, loan size is related to borrower risk, but we find no evidence that borrowers choose it strategically to avoid human screening. Fourth, increases in banks' capital buffers are associated with lower bank rates. These patterns are descriptive and guide the model in Section 4.

3.1 Credit and Risk Allocation Across the Online Application Cap

We take the screening structure documented in Section 2 as given and provide descriptive evidence on how the allocation of credit and risk differs between loans below and above the banks' online application cap, where banks switch from online to human screening while

FIGURE 2: Loan Size Density by Online Application Cap



Note: Density of originated loan amounts for banks whose online application cap is £25,000, banks whose cap is £50,000, and non-banks, with each bank's cap measured at the time of origination. Bank densities contract just above each group's cap.

non-banks continue to screen online.

3.1.1 Credit Allocation

Figure 2 plots the density of originated loan sizes for three groups: loans originated by banks whose online application cap was £25,000 at the time of origination, loans originated by banks whose cap was £50,000, and loans originated by non-banks. Each group of banks concentrates its lending below its cap, and its density contracts sharply just above it, near £25,000 for the first group and near £50,000 for the second.

Motivated by the contraction at each cap, we divide the market into two segments according to how the borrower's bank would have screened each loan. Since more than 93% of bank loans in our data are originated by the borrower's existing main current account provider, we compare each loan with this bank's online application cap in the quarter of origination. Loans within the cap form the *below-cap segment* and loans above it the *above-cap segment*. Non-bank loans are classified in the same way, by comparing their size with the cap of the borrower's bank. As Table 2 shows, banks account for 73% of loans in the below-cap segment, where both lender types screen automatically, and for 19% in the above-cap segment, where banks switch to human screening while non-banks continue to screen automatically.

TABLE 2: Market Shares by Lender Type and Segment

	Below-cap segment	Above-cap segment
Bank	73%	19%
Non-bank	27%	81%
Number of loans	22,479	35,812

Note: Market shares by lender type and segment, computed by loan count within each segment. Each loan is assigned to the below-cap or above-cap segment according to whether its size falls within or above the online application cap that the borrower's main current account provider had in place when the loan was originated. The same rule classifies bank and non-bank loans.

To check that these patterns reflect the cap rather than other differences between banks, we use a case study of one bank in our sample that raised its online application cap from £25,000 to £50,000. Among borrowers whose main current account was with this bank, the bank share of loans between £25,000 and £50,000 rose by about 21 percentage points after the increase relative to smaller loans, with no comparable rise above £50,000 (Appendix B). Bank lending thus follows the cap when it moves, which supports reading the contraction in Figure 2 and the gap in Table 2 as features of the cap.

3.1.2 Risk Allocation

We next compare default across the four pairs of lender type and segment. Besides realised default, we compute for each loan the component of default that is not accounted for by observable borrower characteristics, defined as realised minus predicted default, where predicted default is based on observables alone. Table 3 reports the averages of both measures for each pair.

Below the cap, bank loans default at a rate of 1.9%, compared with 2.6% for non-bank loans. Above the cap, the two lender types diverge: the bank default rate falls to 0.3%, while the non-bank default rate rises slightly to 2.8%. The gap between non-bank and bank default therefore widens from 0.7 to 2.5 percentage points across the two segments.

The gap also widens in the component of default that observables do not explain. Below the cap, bank loans default about 0.5 percentage points less than their observables predict and non-bank loans about 0.1 points more, a gap of 0.6 percentage points. Above the cap, these become 0.8 points less and 0.5 points more, so the gap doubles to 1.3 percentage points. Part of the widening in realised default thus reflects observable characteristics, but

TABLE 3: Default by Lender Type and Segment

	Realised default		Not explained by observables	
	Below cap	Above cap	Below cap	Above cap
Bank	1.88	0.33	−0.53	−0.82
Non-bank	2.58	2.83	0.12	0.46
Non-bank minus bank	0.70	2.50	0.64	1.28

Note: Average one-year default rates in percent. The component not explained by observables is realised minus predicted default, where predicted default is the fitted probability from a probit of default within one year of origination on indicators for young age, below-median assets and credit score categories, and region, sector, quarter and loan maturity fixed effects. Results are nearly identical when the probit uses quartiles of age and assets and deciles of credit score. Segments are defined as in Section 3.1.

the difference between lender types in risk that observables do not capture also grows sharply above the cap.

To check that this widening reflects the cap rather than loan size, we compare loans just below and above £25,000, using borrowers whose bank caps at £50,000 as a placebo group. For borrowers whose bank caps at £25,000, realised bank default falls from 1.8% to 0.2% across the threshold, and unexplained default from −0.8 to −1.9 percentage points. In the placebo group, neither falls: realised default is 1.8% below and 2.1% above the threshold. Non-bank default remains broadly stable (Appendix C). The drop in bank default thus appears only where the cap binds.

Together with Table 2, these patterns show that above the cap, banks originate a much smaller share of loans, and the loans they originate default far less often than non-bank loans, even after accounting for observable borrower characteristics. These patterns are descriptive, and we do not attribute them to the screening channel alone. They motivate a model that allows the bank's screening and its appeal to borrowers to differ between its automated and human channels.

3.2 Pricing and Unobserved Risk

We next ask whether interest rates contain information about borrower risk that we do not observe. We are agnostic about the source of such information (Liberti and Petersen, 2019): it may come from hard information that lenders observe but we do not, from soft information gathered during screening, or from a better ability to process the same informa-

tion. To answer this, we estimate the correlation between interest rates and ex-post default, conditional on observable firm characteristics, separately for each pair of lender type and segment. A positive correlation could also arise if higher rates themselves raise default, through moral hazard or a heavier repayment burden. In Appendix F, we compare default on floating-rate and fixed-rate loans around the post-pandemic rise in interest rates, which raised repayment burdens only for the former, and find no evidence in our data that higher rates increase default. We therefore read a positive conditional correlation as indicating risk that is unobserved to us but correlated with prices.

We find that loans that later default carry higher rates at origination for both lender types and in both segments. We also find that banks charge lower rates than non-banks, and that their rate advantage is largest above the cap, where their market share is smallest.

We estimate variants of the following regression:

$$r_{ijmt} = \alpha_{jm} + \lambda_{jm} \mathbb{1}^{D}_{ijmt} + \Gamma X_{it} + \Phi_{F.E.} + \epsilon_{ijmt}, \tag{1}$$

where r_{ijmt} is the interest rate on a loan originated at time t to firm i by a lender of type $j \in \{\text{bank}, \text{non-bank}\}$ in segment $m \in \{\text{below cap}, \text{above cap}\}$. The intercept α_{jm} captures the rate level of each lender type and segment pair, normalised to zero for non-bank loans below the cap. $\mathbb{1}^{D}_{ijmt}$ equals one if the loan defaults within one year of origination, so λ_{jm} measures how much higher the rate at origination is for loans that later default within pair jm, conditional on observables. X_{it} contains firm age, asset size and credit score, and $\Phi_{F.E.}$ includes region, sector, quarter and loan maturity fixed effects.

Panel A of Table 4 reports rate levels. Relative to non-bank loans in the same segment, bank loans carry rates about 3.1 percentage points lower below the cap and about 6.2 percentage points lower above it, conditional on observables. Comparing with Table 2, the bank share of originated loans is smallest exactly where this rate advantage is largest: banks originate 73% of loans below the cap but only 19% above it. Rates also rise with observable risk, as Panel C shows: young firms, firms with below-median assets and firms in riskier credit score categories pay higher rates.

The larger bank rate advantage above the cap is not simply a feature of larger loans. After the bank in our case study raised its online application cap, its rates on loans between £25,000 and £50,000, which it now screens automatically, rose relative to the rates other

TABLE 4: Interest Rates and Ex-Post Default by Lender Type and Segment

	Categorical controls		Granular controls
	(1)	(2)	(3)
Panel A. Rate level, α_{jm}			
Bank, below cap	-3.125^{***}	-3.092^{***}	-3.154^{***}
	(0.309)	(0.314)	(0.312)
Bank, above cap	-5.508^{***}	-5.475^{***}	-5.342^{***}
	(0.287)	(0.292)	(0.296)
Non-bank, above cap	0.718^{***}	0.735^{***}	0.834^{***}
	(0.229)	(0.234)	(0.240)
Panel B. Default gap, λ_{jm}			
Bank, below cap		1.069^{***}	1.049^{***}
		(0.237)	(0.238)
Bank, above cap		1.504^{***}	1.396^{***}
		(0.421)	(0.407)
Non-bank, below cap		2.031^{***}	2.039^{***}
		(0.402)	(0.399)
Non-bank, above cap		1.081^{***}	1.074^{***}
		(0.135)	(0.137)
Panel C. Firm characteristics			
Age: young	0.291^{***}	0.283^{***}	
	(0.035)	(0.035)	
Assets: below median	0.350^{***}	0.345^{***}	
	(0.060)	(0.060)	
Credit score: normal	0.646^{***}	0.641^{***}	
	(0.053)	(0.053)	
Credit score: risky	0.920^{***}	0.907^{***}	
	(0.065)	(0.065)	
Observations	58,289	58,289	58,289
R^2	0.518	0.520	0.523

Notes: OLS estimates of equation (1). The dependent variable is the annualised interest rate in percentage points. Segments are defined as in Section 3.1. Panel A is relative to non-bank loans below the cap. The omitted firm is old, has above-median assets and is in the lowest-risk credit score category. Granular controls are quartiles of age and assets and deciles of credit score. All columns include quarter, region, sector and maturity fixed effects. Standard errors clustered at the lender × year × region level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

lenders charged on loans of the same size (Appendix B).

Panel B reports the default gap λ_{jm}. In Column (2), loans that default within a year carry higher rates at origination in all four pairs, by about 1.1 percentage points for bank loans below the cap, 1.5 for bank loans above it, 2.0 for non-bank loans below it, and 1.1 for non-bank loans above it. Below the cap, the non-bank gap is significantly larger than the bank gap. Column (3) replaces the categorical controls with quartiles of age and assets and deciles of credit score, and the default gaps are nearly unchanged.

We read these patterns as descriptive. They indicate that interest rates contain information about default risk beyond what our observables capture, for both lender types and in both segments. They do not on their own measure screening precision, since the rate gap is computed only among originated loans and therefore also reflects which borrowers each lender type lends to. The structural model uses them, among other moments, to discipline screening precision for each lender type j and screening channel.

3.3 Loan Size and Borrower Risk

We next ask two questions about loan size: whether it is related to borrower risk beyond what our observables capture, and whether borrowers choose it strategically. For the second, we focus on the form of strategic choice that would matter for comparing the two segments: adjusting loan size to fall below the bank's online cap in a way that is correlated with risk. For the first, we estimate

$$\log(\ell_{ijt}) = \alpha + \beta \mathbb{1}^D_{ijt} + \Gamma X_{it} + \Phi_{F.E.} + \epsilon_{ijt}, \tag{2}$$

where ℓ_{ijt} is the amount of the loan originated at time t to firm i by a lender of type j, $\mathbb{1}^D_{ijt}$ equals one if the loan defaults within one year of origination, and X_{it} and $\Phi_{F.E.}$ are the firm characteristics and fixed effects used in Table 4.

Table 5 reports the results, with coefficients read as percentage differences since the dependent variable is in logs. Loan size rises with observable firm quality: firms with below-median assets take loans about 67% smaller than firms with above-median assets, firms in the safest credit score category take loans about 43% larger than those in the riskiest, and younger firms take loans about 3% smaller than older ones. Conditional on these observables, loans that later default are about 10% larger than loans that do not.

TABLE 5: Loan Size and Ex-Post Default

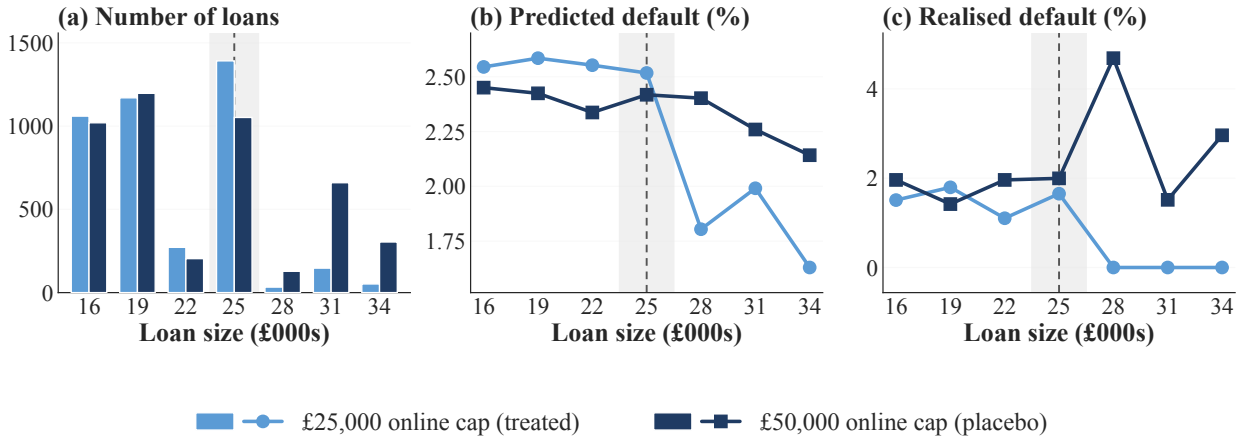
	Log loan amount
$\mathbb{1}^D$	0.094^{***}
	(0.023)
Age: young	-0.026^{***}
	(0.009)
Assets: below median	-1.112^{***}
	(0.028)
Credit score: normal	0.141^{***}
	(0.009)
Credit score: safe	0.358^{***}
	(0.015)
Observations	58,289
R^2	0.510

Notes: OLS estimates of equation (2). The dependent variable is the log of the loan amount. $\mathbb{1}^D$ equals one if the loan defaults within one year of origination. The omitted firm is old, has above-median assets and is in the riskiest credit score category. Includes quarter, region, sector and maturity fixed effects. Standard errors clustered at the lender × year × region level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

To address the second question, we compare borrowers whose bank caps online applications at £25,000 with a placebo group of borrowers whose bank caps them at £50,000. If borrowers reduced their loans to stay in the automated channel, the first group would bunch more just below £25,000, and the bunching would be concentrated among riskier borrowers, who have the most to lose from human screening. Figure 3 shows no such pattern. Loan sizes bunch at £25,000 in both groups, consistent with £25,000 being a common round amount, and somewhat more in the treated group. But the borrowers who bunch are not riskier: for the treated group, both the default rate predicted by observables and the actual default rate at and just below the cap are similar to those of slightly smaller loans and to those of the placebo group. Above the cap, both measures fall for the treated group, whose loans there are screened through relationship managers. We therefore find no evidence that borrowers sort around the cap in a way that is correlated with their observable quality or realised risk.

Together, these patterns indicate that loan size varies with observable firm characteristics and is correlated with borrower risk beyond them, while we find no evidence of sorting around the cap that is correlated with risk. The structural model accordingly treats loan size as

FIGURE 3: Loan Size Around £25,000 by Online Application Cap



Notes: Loans to borrowers whose main current account provider capped online applications at £25,000 (treated) or £50,000 (placebo) in the quarter of origination, in £3,000 loan size bins. Panel (a) shows the number of loans, panel (b) the default rate predicted by observable firm characteristics, and panel (c) the actual one-year default rate. The shaded area marks the £25,000 bin.

determined by both observed and unobserved borrower characteristics, rather than chosen strategically.

3.4 Bank Pricing and Capital Buffers

Finally, we ask whether bank rates co-move with how tightly capital regulation binds. We measure this with the bank's capital buffer, the difference between its capital ratio and its regulatory minimum: a smaller buffer means the bank is closer to its requirement and more constrained by regulation. We estimate

$$r_{ibt} = \beta\,\Delta\text{Buffer}_{bt} + \sum_{k=1}^{2} \beta_k\,\Delta\text{Buffer}_{b,t-k} + \psi\,\Delta\text{Assets}_{bt} + \Gamma X_{it} + \mu_b + \Phi_{F.E.} + \epsilon_{ibt}, \quad (3)$$

where r_{ibt} is the interest rate on a loan originated by bank b to firm i in quarter t, ΔBuffer_{bt} is the change in bank b's capital buffer in quarter t, in percentage points, ΔAssets_{bt} is the change in its total assets, X_{it} are the firm characteristics used in Table 4, μ_b are bank fixed effects, and $\Phi_{F.E.}$ includes region, sector, quarter and maturity fixed effects. The sample contains bank loans only.

The change in the buffer varies at the bank-quarter level. Bank fixed effects absorb

TABLE 6: Capital Buffer Changes and Bank Rates

	(1)	(2)	(3)	(4)
Δ Capital buffer$_{bt}$	-0.041^{*}	-0.052^{**}	-0.083^{***}	-0.082^{***}
	(0.023)	(0.024)	(0.030)	(0.029)
Firm controls	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes
Δ Assets$_{bt}$	No	Yes	Yes	Yes
Δ Capital buffer$_{b,t-1}$, Δ Capital buffer$_{b,t-2}$	No	No	Yes	Yes
Region, sector, quarter, maturity FE	Yes	Yes	Yes	No
Quarter $\times$ lender $\times$ year $\times$ region FE, maturity FE	No	No	No	Yes
Observations	23,307	23,307	23,307	21,302

Notes: OLS estimates of equation (3) on bank loans. The dependent variable is the annualised interest rate in percentage points. The capital buffer is the bank's capital ratio minus its regulatory minimum, in percentage points. Firm controls are those of Table 4. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

persistent differences in rate levels across banks, and quarter fixed effects absorb changes common to all banks in a given quarter, such as shifts in monetary policy or in aggregate credit demand. The coefficient β is therefore identified from short-run variation within banks: it compares the rates a bank charges observably similar firms in quarters when its buffer changes.

Table 6 shows that the coefficient on the change in the buffer is negative in all four specifications. It becomes larger in magnitude and significant at the 1% level once we control for lagged buffer changes and, in Column (4), compare loans within the same quarter, region and sector. In Column (4), a one percentage point increase in a bank's buffer is associated with rates about 8 basis points lower on its new loans, a correlation consistent with banks charging higher rates when regulation binds more tightly.

4 Model

We develop a static model of the UK market for unsecured SME loans, built around the facts in Section 3. One bank and one non-bank compete for borrowers with differentiated demand, setting an individual rate for each borrower based on their own screening of its risk. Lenders differ along two dimensions. The first is screening: the precision of screening differs across lenders and, for the bank, between online and human screening. The second is regulation:

the bank bears a cost of capital regulation that rises with the risk weight of each loan. The model captures the link between the two, since the risk weight depends on the bank's own assessment of the borrower's risk, so the bank's screening determines its regulatory cost.

Modelling one lender of each type keeps the model tractable, and two features of the data support this choice. First, pricing and default patterns are broadly similar across lenders of the same type, so a single lender per type captures the main differences in the market. Second, bank choice is largely predetermined: more than 93% of bank loans are originated by the borrower's existing main current account provider. Each borrower therefore effectively faces a single bank, whose online application cap we take as given.

4.1 Borrowers

Borrowers, indexed by i, differ in observable characteristics X_i and in unobserved risk θ_i. Lenders do not observe i and learn about it only through screening, which is the source of asymmetric information in the model. A borrower can take an unsecured term loan from a lender $j \in \mathcal{J} = \{B, N\}$, where B is its bank, the provider of its main current account, and N is the non-bank, or choose the outside option $j = 0$ of not borrowing, so that its full choice set is $\{0\} \cup \mathcal{J}$.

Default. A borrower that takes a loan from lender j defaults when a latent default index exceeds a threshold U_0^F:

$$U_{ij}^F = \beta^F X_i + \beta_\theta^F \theta_i + \delta_j^F + \epsilon_{ij}^F, \tag{4}$$

where ϵ_{ij}^F is a standard normal shock realised after the loan is originated. The index is a reduced-form measure of how likely the borrower is to default. Observable characteristics X_i and unobserved risk θ_i capture the firm's ability to generate cash flows to repay, as known at origination, the shock ϵ_{ij}^F captures cash-flow shocks realised after origination, and δ_j^F captures differences across lenders in the borrower's incentive to repay. The probability of default is therefore

$$PD_{ij}(X_i, \theta_i) = \Phi\left(\beta^F X_i + \beta_\theta^F \theta_i + \delta_j^F - U_0^F\right). \tag{5}$$

The default index does not depend on the interest rate r_{ij}: rates affect which borrowers take a loan from each lender, but not whether a given borrower defaults. This is consistent with the evidence in Appendix F, where we find no evidence that higher repayment burdens increase

default. The term δ_j^F shifts default by lender type, capturing differences in repayment incentives, such as the strength of contract enforcement or the value to the borrower of preserving its banking relationship. It is motivated by the evidence in Appendix Table D.1 that firms borrowing from both lender types are more likely to default on their non-bank loan.

Loan size. Each borrower seeks a loan of size ℓ_i, determined by its characteristics:

$$\log(\ell_i) = \beta^\ell X_i + \beta_\theta^\ell \theta_i + \epsilon_i^\ell, \tag{6}$$

where ϵ_i^ℓ is an idiosyncratic shock to credit demand. The term $\beta^\ell X_i$ captures how funding needs vary with observable characteristics such as firm size, and $\beta_\theta^\ell \theta_i$ allows loan size to be correlated with unobserved risk, consistent with the evidence in Section 3.3.

We assume the interest rate is the only margin on which lenders compete and screen (Stiglitz and Weiss, 1981; Crawford, Pavanini and Schivardi, 2018). We therefore treat ℓ_i as the amount the borrower requests before meeting any lender, and the loan is originated at this amount. Loan size then feeds into screening: since it depends on the borrower's unobserved risk, lenders use it to learn about that risk (Section 4.2). Lenders quote individual rates for the amount requested.[14] Taking loan size as given keeps the lender's problem one-dimensional and matches our data, which record originated amounts but not applications or counteroffers.

Segment. Loan size determines the borrower's segment. Let $\bar{\ell}_i$ denote the online application cap of borrower i's bank at the time of borrowing. The borrower's segment is

$$m(i) = \begin{cases} \text{below cap} & \text{if } \ell_i \leq \bar{\ell}_i, \\ \text{above cap} & \text{if } \ell_i > \bar{\ell}_i. \end{cases} \tag{7}$$

The segment determines how the bank screens the borrower: with automated screening below the cap and with human screening above it, in the sense defined in Section 2. The non-bank

[14] In discrete-continuous choice models, borrowers know the rate attached to each loan amount before choosing, for example from a posted menu, so the optimal amount can be derived from indirect utility via Roy's identity (Train, 1986). In our setting, a borrower would instead need to anticipate how each lender would price every possible amount, including what each amount would signal about its risk. We abstract from this choice and take the requested amount as given.

screens automatically in both segments.

Lender choice. Within its segment $m = m(i)$, the borrower chooses among the offers it receives. The indirect utility of borrower i from a loan from lender j at rate r_{ijm} is

$$U^D_{ijm} = \underbrace{-\alpha r_{ijm} + \beta^D X_i + \delta^D_{jm}}_{V_{ijm}} + \epsilon^D_{ij}, \tag{8}$$

where α measures price sensitivity, the shock ϵ^D_{ij} is type I extreme value, and δ^D_{jm} captures non-price features of lender j in segment m, such as the speed and convenience of the application:

$$\delta^D_{jm} = \begin{cases} \delta^D_N & \text{if } j = N, \\ \delta^D_B & \text{if } j = B \text{ and } m = \text{below cap}, \\ \delta^D_B - \delta^D_h & \text{if } j = B \text{ and } m = \text{above cap}. \end{cases} \tag{9}$$

The non-bank takes applications online in both segments, so its non-price features do not vary with the segment. For the bank, δ^D_h captures the change in the borrower's valuation when the application moves from online to a relationship manager above the cap: a positive value reflects the additional time and effort this involves.

Our data record originated loans only. We therefore assume that both lenders are in every borrower's choice set: each lender makes an offer to every borrower (Crawford, Pavanini and Schivardi, 2018). Lenders screen through the rates they offer rather than through approval. A lender that assesses a borrower as very risky offers a high rate, which leads the borrower to choose the rival lender or not to borrow. A high rate thus plays the role of a rejection. Each borrower therefore chooses from $\{0\} \cup \mathcal{J}$. Normalising $U^D_{i0} = \epsilon^D_{i0}$, the probability that borrower i chooses lender $j \in \mathcal{J}$ is

$$Q_{ijm} = \frac{e^{V_{ijm}}}{1 + \sum_{k \in \mathcal{J}} e^{V_{ikm}}}. \tag{10}$$

4.2 Information Structure

Lenders do not observe borrower risk θ_i and must infer it when setting rates. They draw on two sources of information: the requested loan size, which is observed by all lenders, and a private signal obtained through screening. Each lender combines these sources by Bayesian

updating.

Public signal. Because loan size depends on θ_i through equation (6), the requested amount is informative about risk conditional on observables. Inverting the loan size equation yields a public signal:

$$\tilde{s}_i \equiv \frac{\log(\ell_i) - \beta^\ell X_i}{\beta^\ell_\theta} = \theta_i + \tilde{\epsilon}_i, \qquad \tilde{\epsilon}_i = \frac{\epsilon^\ell_i}{\beta^\ell_\theta}, \tag{11}$$

with precision $\tilde{\tau}$, which increases with the strength of the relationship between loan size and risk relative to the variance of the demand shock.

Private signals. Each lender additionally obtains a private signal through screening:

$$s_{iN} = \theta_i + \epsilon_{iN}, \quad \epsilon_{iN} \sim \mathcal{N}\big(0, (\tau^N)^{-1}\big), \qquad s_{iBm} = \theta_i + \epsilon_{iBm}, \quad \epsilon_{iBm} \sim \mathcal{N}\big(0, (\tau^B_m)^{-1}\big). \tag{12}$$

The precision of the signal represents the lender's screening ability. We remain agnostic about the nature of the underlying information, which may be hard or soft. Since the non-bank screens automatically in both segments, it has a single precision τ^N. The bank's precision depends on its screening channel, τ^B_m with $m \in \{\text{below cap}, \text{above cap}\}$, reflecting its switch from automated to human screening at the cap.

Posterior beliefs. Lender j forms its posterior about θ_i by Bayesian updating on its information set $\mathcal{I}_{ijm} = \{\tilde{s}_i, s_{ijm}\}$, where s_{ijm} is s_{iN} for the non-bank in both segments and s_{iBm} for the bank. The informativeness of this set therefore depends on the lender's screening precision: τ^N for the non-bank, and τ^B_m for the bank in segment m. Lenders price loans based on the default probability implied by this posterior (Section 4.3).

4.3 Lenders

The bank and the non-bank compete on interest rates. Each lender observes the borrower's characteristics X_i, its loan size ℓ_i and its own private signal, but not the rival's signal, and sets a rate for each borrower to maximise expected profit given its information set $\mathcal{I}_{ijm}$. We focus on Bayesian Nash equilibria in pricing strategies, in which each lender's rate is a best response to the rival's pricing strategy.

Expected profit. Lender j's expected profit from borrower i in segment m is

$$\mathbb{E}\big[\Pi_{ijm} \mid \mathcal{I}_{ijm}\big] = \mathbb{E}\Big[Q_{ijm}(r_{ijm}, r_{ikm})\big((1 - PD_{ij})\, r_{ijm} - PD_{ij}\, LGD_j - c_{ijm}\big) \;\Big|\; \mathcal{I}_{ijm}\Big], \quad (13)$$

where Q_{ijm} is the demand probability in equation (10), k denotes the rival lender, PD_{ij} is the default probability in equation (5), LGD_j is the lender's loss given default, and c_{ijm} is its marginal cost. Since X_i is observed by both lenders, the expectation is taken over the borrower's risk θ_i and the rival's private signal, using lender j's posterior. The rival's signal matters because the rival's rate r_{ikm} depends on it: a lender is more likely to win a borrower when the rival has priced it high, that is, when the rival's signal indicates high risk. Expected profit therefore accounts for a winner's curse, since winning a borrower is itself informative about its risk.

Costs. The two lenders differ in their marginal costs:

$$c_{iNm} = c_N, \qquad c_{iBm} = c_B + \kappa\, \rho^B RW_{iBm}, \quad (14)$$

where c_N and c_B capture funding and origination costs. For the bank, the additional term captures the shadow cost of capital regulation: each loan requires the bank to hold capital equal to the minimum capital ratio ρ^B times the loan's risk weight RW_{iBm}, and κ is the shadow cost of each unit of this required capital. A higher capital ratio or a higher risk weight therefore raises the bank's cost of lending.

The risk weight links screening and regulation. It depends on the bank's assessment of the borrower's default risk, $RW_{iBm} = RW\big(\mathbb{E}[PD_{iB} \mid \mathcal{I}_{iBm}]\big)$, which is formed from its screening signal. The bank's screening therefore determines not only how it prices default risk but also how much regulatory capital each loan requires.

Optimal rates. The first-order condition of equation (13) with respect to r_{ijm} implies that the optimal rate is the sum of an effective marginal cost and an effective markup:

$$r^*_{ijm} = \underbrace{\frac{\mathbb{E}\big[Q'_{ijm}\,(c_{ijm} + PD_{ij}\, LGD_j) \mid \mathcal{I}_{ijm}\big]}{\mathbb{E}\big[Q'_{ijm}\,(1 - PD_{ij}) \mid \mathcal{I}_{ijm}\big]}}_{\text{effective marginal cost}} \underbrace{- \frac{\mathbb{E}\big[Q_{ijm}\,(1 - PD_{ij}) \mid \mathcal{I}_{ijm}\big]}{\mathbb{E}\big[Q'_{ijm}\,(1 - PD_{ij}) \mid \mathcal{I}_{ijm}\big]}}_{\text{effective markup}}, \quad (15)$$

where $Q'_{ijm} = \partial Q_{ijm}/\partial r_{ijm} < 0$. The effective marginal cost combines the cost of the loan with its expected loss from default, adjusted for the probability of repayment. The effective markup reflects the lender's market power: it depends on the demand probability relative to its sensitivity to the rate.

5 Estimation and Identification

The model does not admit a closed-form solution: the optimal rate in equation (15) depends on demand and its derivative, which depend on both the lender's own rate and the rival's, so the two lenders' pricing strategies must be solved jointly as a fixed point. We therefore solve the model numerically and estimate its parameters by indirect inference (Gourieroux, Monfort and Renault, 1993), matching moments of the data, including the coefficients of auxiliary regressions, to their counterparts in model simulations. We first describe our parametric assumptions and the sources of identification, then the estimation procedure, and finally the estimates and the model's fit.

5.1 Parametric Assumptions and Calibrated Parameters

We normalise unobserved risk to $\theta_i \sim \mathcal{N}(0,1)$. The loan size shock ϵ_i^ℓ, the default shock ϵ_{ij}^F and the screening noise ϵ_{ijm} are normally distributed and mutually independent, and the demand shocks ϵ_{ij}^D are type I extreme value. Observable characteristics X_i consist of indicators for young age, below-median assets and credit score categories.

We calibrate three sets of parameters outside the model. The bank's minimum capital ratio ρ^B is set to 12.98%, the average minimum CET1 requirement of the banks in our sample, including supervisory add-ons, weighted by their lending. Losses given default are set to $LGD_B = 68.27\%$ and $LGD_N = 80\%$.[15] The risk weight RW_{iBm} follows the Basel internal ratings-based formula for retail exposures, including the supporting factor for SME exposures, evaluated at the bank's assessed default probability $\mathbb{E}[PD_{iB} \mid \mathcal{I}_{iBm}]$ and LGD_B (Appendix A).

[15] The bank value is computed from our regulatory data on banks' capital requirements for SME exposures. The non-bank value is based on publicly available securitisation and recovery rate documentation for a large UK non-bank SME lender, which we do not name for confidentiality reasons.

5.2 Identification

We estimate the model by matching the coefficients of pricing, default and loan size regressions, together with market shares and default rates by lender type and segment, which we compute in the same way in the data and in the model (Section 5.3). Since all parameters are estimated jointly, each one affects many moments at once. Identification nonetheless relies on the fact that each parameter affects some moments in a distinctive way. When two parameters move the same moments in the same direction, we separate them using a moment on which they have different effects.

Some parameters are closely tied to specific moments. The loan size parameters β^ℓ are identified mainly from how loan size varies with observable characteristics, and the default parameters β^F and U_0^F from default rates and from how default varies with observables. The non-price terms δ_N^D and δ_B^D and the parameters β^D are identified mainly from market shares by lender type and borrower characteristics, including the share of potential borrowers who do not borrow. Given the calibrated losses given default, the origination costs c_N and c_B are identified from the level of rates, net of markups and expected default losses. The price sensitivity α is identified from a change in the funding structure of one of the large non-banks in our sample, which stopped raising funds from retail investors during our sample period. This change raised the non-bank's funding costs and led to a persistent widening of the gap between bank and non-bank rates. Since it affects borrowers only through the rates they are offered, the resulting shift in market shares between banks and non-banks across the two periods identifies how strongly borrowers respond to rate differences. We allow non-bank lender's cost to differ between the periods before and after the change, and target market shares and the bank-non-bank rate gap separately in each period.

The remaining parameters govern unobserved risk and how lenders learn about it, and are identified jointly. The weight of unobserved risk in default, β_θ^F, and in loan size, β_θ^ℓ, are disciplined by the extent to which default varies beyond what observables predict, and by the coefficient on ex-post default in the loan size regression, which is positive only if loan size and default share a common unobserved component. The screening precisions are disciplined by two moments in each pair of lender type and segment, both measured conditional on observables: the co-movement of rates with the part of ex-post default that is orthogonal to observable characteristics, and default conditional on observables. These moments reflect both precision and selection. Precision determines the rates a lender offers,

since the more precise its signal, the more closely its offered rates respond to risk that borrower characteristics do not reveal. Borrowers then choose between the two offers, so the loans each lender originates depend on its rates relative to the rival's. The model captures this selection explicitly through the demand system, so the precisions are estimated such that the offered rates, once filtered through borrowers' choices, reproduce these moments among originated loans. The lender default shifter is identified from the difference in default between bank and non-bank loans conditional on observables, net of the selection the model implies.

The bank's costs are separated from its precision and from each other by how they affect bank rates. The shadow cost of capital κ and the origination cost c_B both raise bank rates, but only the capital charge rises with the risk weight, so κ is disciplined by how bank rates vary across groups of borrowers with different observable risk, which determines their risk weights, and c_B by the level of bank rates. Precision and κ both affect how bank rates co-move with default, but differ in how the bank prices borrowers with identical observables. Higher precision tilts the rate schedule among such borrowers, raising rates for those the bank's signal identifies as risky and lowering them for those it identifies as safe, whereas a higher shadow cost of capital raises rates for all of them, by more for those assessed as riskier. Precision therefore shows up mainly in the co-movement of bank rates with unobserved risk, and κ mainly in how bank rates vary with observable risk.

Finally, δ_h^D is disciplined by the change in the bank's market share above the cap relative to the change in its rates. Banks lose market share above the cap although their rates fall relative to non-bank rates, which the model attributes partly to borrowers' valuation of applying through a relationship manager and partly to the selection induced by the bank's pricing.

5.3 Estimation Procedure

For each candidate parameter vector Θ, we solve numerically for the equilibrium pricing strategies, simulate a population of potential borrowers, and compute the moments on the loans they originate. Since we observe only originated loans, each moment is computed on simulated originated loans in the same way as in the data, with the same segment definitions and regression specifications. We choose Θ to minimise the weighted distance between the moments in the data and in the simulations. Appendix G shows numerically that the

TABLE 7: Parameter Estimates

		Estimate	Standard error
Screening precision			
Bank, automated	τ^B_{below}	0.143	0.053
Bank, human	τ^B_{above}	0.605	0.250
Non-bank	τ^N	0.394	0.073
Costs			
Origination cost, bank (%)	c_B	2.71	0.21
Origination cost, non-bank (%)	c_N	6.46	0.56
Shadow cost of capital	κ	0.203	0.103
Demand			
Price sensitivity	α	37.476	2.469
Human channel friction	δ^D_h	3.422	0.219

Note: Indirect inference estimates. Screening precisions are the inverse of the variance of each lender's signal noise, with standard errors computed by the delta method from the estimated standard deviations. The human channel friction enters bank demand above the cap with a negative sign, as in equation (9). Table H.1 in Appendix H reports the remaining parameters.

equilibrium of the pricing game at the estimated parameters is locally unique and robust to perturbations.

5.4 Estimates

Table 7 reports the estimated parameters. We focus on the parameters that govern screening and regulation, and report the remaining parameters in the table H.1.

Screening. Each precision is the inverse of the variance of the noise in a lender's private signal, so that a higher value means the signal tracks borrower risk more closely. As discussed in Section 5.2, the precisions are pinned down mainly by the co-movement of rates with the component of default not explained by observables, and by default conditional on observables, in each pair of lender type and segment. The estimates show that the ranking of lenders depends on the type of information on which their assessment mainly relies. Below the cap, where both lenders screen online and rely primarily on hard information, the non-bank screens more precisely than the bank, with precisions of 0.39 and 0.14. Above the cap, where the bank's assessment is led by a relationship manager who can also draw on soft information,

TABLE 8: Decomposition of Marginal Costs

	Bank	Non-bank
Origination cost	2.71	6.46
Capital shadow cost	1.20	
Marginal cost	3.91	6.46
Expected default loss	0.93	2.14
Effective marginal cost	4.85	8.60

Note: Averages across originated loans at the estimated parameters, in percent. The origination cost captures all costs other than the capital charge and expected default losses. The capital shadow cost is the bank's regulatory capital charge. Expected default loss is the difference between effective and plain marginal cost, reflecting the adjustment for default in equation (15).

the ranking reverses: the bank's precision rises to 0.60, above that of the non-bank. Loan size, by contrast, carries little information about risk, with a precision of 0.012. Table 7 reports standard errors.

Costs and regulation. The bank has a cost advantage over the non-bank, including after accounting for capital regulation: its marginal cost, including the capital shadow cost, is 3.91%, compared with 6.46% for the non-bank, and its effective marginal cost, which adds expected default losses, is 4.85%, compared with 8.60%. Table 8 decomposes these costs. The origination cost is 2.71% for the bank and 6.46% for the non-bank. This component captures all costs of lending other than the capital charge and expected default losses, including differences in funding costs, such as the bank's access to deposits, in operating costs, and in any regulation beyond capital requirements. The bank additionally bears a capital shadow cost of 1.20%, about 31% of its marginal cost, which narrows the cost difference between the two lenders without reversing it. Expected default losses are 0.93% for the bank and 2.14% for the non-bank, reflecting both the default risk of each lender's borrowers and its loss given default.

Demand and markups. The average own-rate elasticity of demand is 2.62: a 10% increase in a lender's rate, reduces its demand by about 26% on average. Effective markups differ across the cap in opposite directions for the two lenders (Table 9). The bank's markup falls from 5.36% below the cap to 2.99% above it, while the non-bank's rises from 4.21%

TABLE 9: Effective Markups by Lender Type and Segment

	Bank	Non-bank
Below cap	5.36	4.21
Above cap	2.99	4.81

Note: Average effective markups across originated loans at the estimated parameters, in percent.

to 4.81%. In the model, this pattern mainly reflects δ_h^D, which lowers the value of a bank loan above the cap for a given rate: the bank's margin falls, and the non-bank, facing a less attractive rival, earns a higher one.

Although δ_h^D enters the model as borrowers' valuation of applying through a relationship manager, it is identified from the bank's market share above the cap relative to its rates. The data point to a friction of this kind. As Section 3.2 shows, the rate gap between banks and non-banks widens above the cap, conditional on observables and ex-post default, while the bank's market share falls. If borrowers responded only to rates, a bank that becomes relatively cheaper would gain market share. Part of the decline reflects the selection induced by the bank's more precise screening, and the remainder is attributed to δ_h^D. The parameter therefore captures any friction in the bank's human channel that makes borrowers less likely to take a bank loan above the cap without being reflected in its rate, including the time and effort of a relationship manager application, non-price loan terms, and bank-side constraints that operate through quantities rather than prices, such as limited relationship manager capacity. We interpret it accordingly, as a composite of non-price frictions in the bank's human channel.[16]

5.5 Model Fit

Table 10 compares selected targeted moments in the data and in the model; Table I.1 in Appendix I reports all targeted moments. The model reproduces the rate levels of each lender type and segment, including the larger bank rate advantage above the cap, and

[16]We model these frictions with a single demand-side term for two reasons. First, our data record originated loans but not applications, non-price terms or relationship manager capacity, so their separate contributions are not identified; a single term summarises their joint effect without imposing structure the data cannot discipline. Second, all of these frictions arise from the human channel, and would therefore be removed together by the counterfactual in which the bank screens automatically above the cap. For this counterfactual, the relevant object is their combined effect, which is what δ_h^D measures.

TABLE 10: Model Fit: Selected Targeted Moments

	Data	Model
Rate level relative to non-bank, below cap		
Bank, below cap	−3.42	−3.82
Bank, above cap	−6.09	−6.85
Non-bank, above cap	0.88	1.14
Default gap in rates, conditional on observables		
Bank, below cap	1.03	0.78
Bank, above cap	1.40	1.86
Non-bank, below cap	2.05	1.49
Non-bank, above cap	1.06	1.35
Default rate (%)		
Bank, below cap	1.87	1.77
Bank, above cap	0.32	0.61
Non-bank, below cap	2.57	2.90
Non-bank, above cap	2.81	2.47
Market share		
Non-bank, below cap	0.27	0.28
Non-bank, above cap	0.81	0.77

Note: Rate levels and default gaps are coefficients of the pricing regression in Table 4, in percentage points; the default gap is the difference in rates at origination between loans that later default and loans that do not, conditional on firm characteristics. Market shares are shares of originated loans within each segment.

the non-bank's market share in both segments. It also reproduces the changes within banks across the cap that are central to our mechanism: when banks switch to human screening, the default rate of their loans falls and the co-movement of their rates with default, conditional on observables, strengthens, as in the data. These are the moments that, as discussed in Section 5.2, discipline the bank's screening precision in each channel, so matching their change across the cap supports the estimated difference between banks' automated and human screening.

5.6 Screening Precision: A Numerical Illustration

This subsection uses a numerical exercise to show how a lender's screening precision shapes its rates, its borrower pool and its profit. For the bank, precision also determines the risk-weighted capital it must hold. We vary the bank's precision, holding all other parameters at

their estimated values, including the non-bank's precision, and solve for the new equilibrium at each level. For illustration, we focus on one group of borrowers, young firms with below-median assets in the normal credit score category, in the segment where both lenders screen online. The patterns are similar for other groups.

Figure 4 shows how precision shapes the bank's pricing. A more precise bank offers a steeper rate schedule in its private signal. This is because its private signal is more informative about unobserved risk, so its risk assessment puts more weight on it, and its offered rates respond more strongly to it.

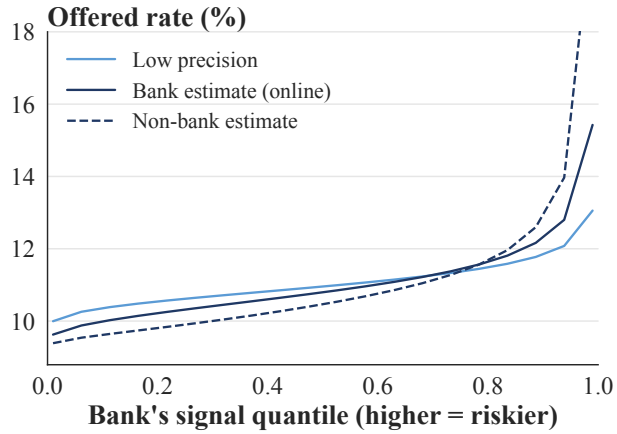
How does this steeper pricing affect the bank? It changes which borrowers the bank lends to. Figure 5 shows three consequences. First, the default rate on the bank's loans falls as its precision rises (panel a). Since observable characteristics are held fixed, this drop reflects only a change in the unobserved risk of the borrowers the bank retains. Second, the gap between the rates on defaulted and repaid loans widens (panel b). The bank's rates thus track risk more closely, even as its borrowers become safer in unobserved risk. Third, panel (c) displays the bank's ex-ante expected profit (i.e., the expected profit in equation (13) integrated over the distribution of its private signal, before any signal is realised). Profit rises steadily with precision, which measures the value of precision to the bank: raising its precision from its estimated online level to that of the non-bank increases its profit by about 12 percent.

Finally, the estimated model highlights the link between screening and regulation at the centre of our paper. A more precise bank retains borrowers with lower risk weights, and hence faces a lower capital cost. Raising the bank's precision from its estimated online level to that of the non-bank lowers its average capital cost per loan from 152 to 119 basis points (i.e., by about 21 percent).

6 Counterfactuals

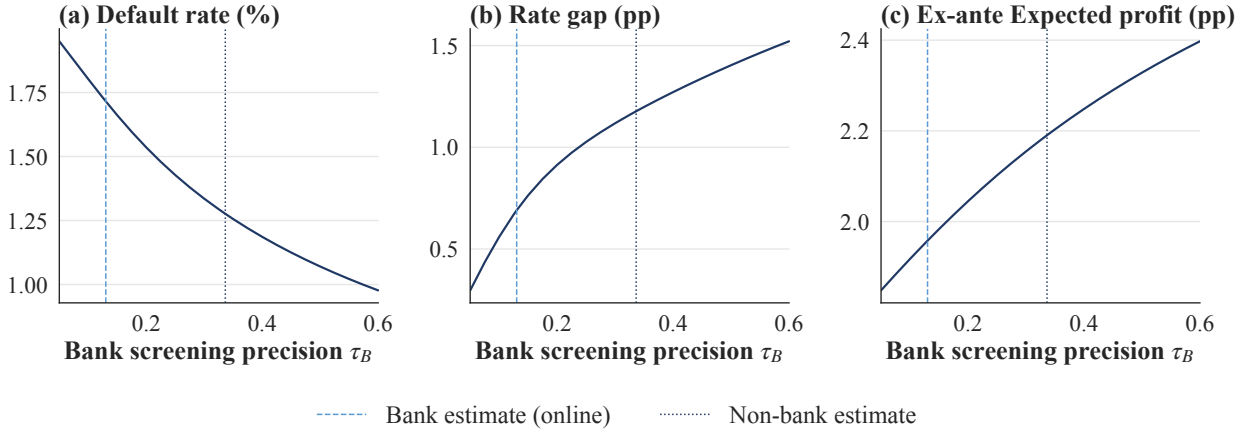
We use the estimated model to study the two features of the market at the centre of the paper: the way banks screen borrowers and the regulation of banks' capital. In the first counterfactual, the bank moves its screening above the cap online, so that it screens all loans as it currently does below the cap. This lowers the bank's precision above the cap to its online level and removes the non-price friction of human screening captured by δ_h^D. It asks

FIGURE 4: The Bank's Offered Rate by Screening Precision



Note: The bank's offered rate below the cap, by quantile of its private signal, for young firms with below-median assets in the normal credit score category. Lines show the bank's estimated online precision (0.14), the non-bank's estimated precision (0.39) and a low precision of 0.05. All other parameters are held at their estimated values, and both lenders' pricing is re-solved at each precision.

FIGURE 5: Screening Precision, Default and Profit



Note: Equilibrium outcomes below the cap for young firms with below-median assets in the normal credit score category, as the bank's screening precision varies. All other parameters are held at their estimated values, including the non-bank's precision. Panel (a) shows the default rate on the bank's originated loans. Panel (b) shows the difference between the average rate on originated loans that later default and on those that repay. Panel (c) shows the bank's ex-ante expected profit, its expected profit per unit lent before it observes its signal, averaged over all potential borrowers in the group. Dashed and dotted lines mark the bank's estimated online precision and the non-bank's estimated precision.

how moving screening online changes credit and risk allocation, and, since the bank's risk assessment determines its risk weights, its regulatory cost. In the second counterfactual, we raise the bank's minimum capital requirement, which increases the regulatory cost of each loan in proportion to its risk weight, and hence to the bank's assessment of the borrower's risk. This asks how capital regulation affects the rates banks charge, the allocation of borrowers between banks and non-banks, and the risk each lender bears, and in particular whether its effects differ between the segment where the bank screens online and the segment where it screens through relationship managers. In both counterfactuals, we hold all other parameters at their estimated values and solve for the new equilibrium.

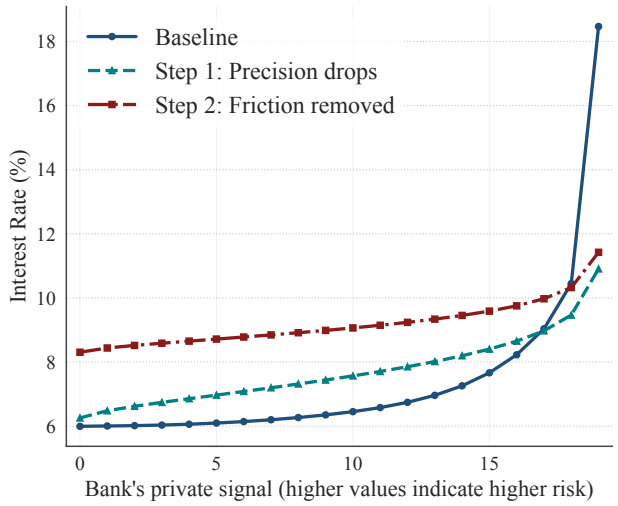
6.1 Online Screening for All Bank Loans

The banking sector is moving from branches to online channels, as banks close branches and increasingly rely on technology in place of staff.[17] Our first counterfactual asks what would happen if the bank screened all loans online. We consider a bank that replaces its relationship managers above the cap with its own online screening, so that it screens all loans as it currently does below the cap. The bank therefore screens more like the non-bank, which screens online at every loan size. Moving the bank's screening online changes two things above the cap: its precision falls to its online level, and the frictions of applying through a relationship manager, captured by δ_h^D, disappear. We introduce these changes one at a time, first lowering precision and then removing the friction. Table 11 and Figure 6 report the results above the cap.

Lower precision makes the bank's rates less responsive to its signal, the reverse of the pattern in Section 5.6. With a noisier signal, the bank prices borrowers more alike: in Figure 6, its rate schedule becomes less convex in its private signal, with lower rates for the borrowers it assesses as riskiest and higher rates for the others. Risky borrowers now receive lower offers from the bank, so it retains a riskier pool. Expected default losses on its loans rise by 40.0 basis points. The capital cost of its loans also rises by 41.4 basis points, which reflects the link between screening and regulation: the bank's risk weights rest on its own assessment, which is now noisier and applies to a riskier pool.

[17] For example, a branch closure tracker by Which?, the UK's largest independent consumer organisation, records bank branch closures since 2015, and a BBC News article reports that Barclays is cutting jobs as it relies more on artificial intelligence.

FIGURE 6: Bank's Offered Rates Above the Cap Under Online Screening



Note: Rate offered by the bank above the cap as a function of its private signal, for young firms with below-median assets in the normal credit score category. Step 1 sets the bank's precision above the cap to its online level; step 2 additionally sets δ_h^D to zero.

Removing δ_h^D eliminates the non-price frictions of borrowing from the bank above the cap, where human screening is in place. Since δ_h^D captures the full set of these frictions in the bank's human channel (Section 5.4), this accounts for the size of the effects. The difference between the two columns of Table 11 gives the effect of removing the friction. The bank's markup rises by a further 226.8 basis points, which shifts its rate schedule up in Figure 6. The non-bank, now competing with a bank without these frictions, lowers its rates by 89.7 basis points.

In total, the rates on the bank's loans above the cap rise by 314.8 basis points. The capital cost and expected default losses account for 48.4 and 49.0 basis points of the increase, and the markup for the remaining 217.4. The default rate of its loans rises by 0.97 percentage points, most of it from the lower precision.

Moving screening online also expands credit. The share of firms that borrow above the cap rises by 25.2 percentage points, mostly from removing the non-price friction of human screening. In the model, this expansion, together with the higher markup, raises the bank's expected profit. That banks nevertheless screen larger loans through relationship managers suggests that moving screening online involves costs or constraints outside our model. In our data, one bank moved in this direction, raising its online application cap

TABLE 11: Counterfactual: Moving the Bank's Screening Online Above the Cap

	Change relative to baseline	
	Lower precision	Lower precision and no friction
Bank		
Rate on originated loans	72.1	314.8
Decomposition:		
Capital cost	41.4	48.4
Expected default loss	40.0	49.0
Markup	−9.3	217.4
Default rate (pp)	0.65	0.97
Non-bank		
Rate on originated loans	7.8	−82.0
Decomposition:		
Expected default loss	6.5	−9.7
Markup	1.2	−72.3

Note: Changes relative to the baseline equilibrium above the cap, in basis points unless stated otherwise. The first column sets the bank's precision above the cap to its online level; the second additionally sets δ_h^D to zero. The difference between the two columns is the effect of removing the friction. Outcomes below the cap are unchanged.

from £25,000 to £50,000 in 2023 (Appendix B). The counterfactual treats the switch as costless, whereas screening larger loans online may require developing and validating new credit models. Banks' use of human screening may also reflect supervisory expectations of human oversight of credit decisions, which serve prudential objectives, such as limiting model risk, that our model does not capture (European Banking Authority, 2020).[18] Non-banks are not subject to the same prudential supervision, so these expectations apply mainly to banks. The profit gain in the counterfactual is therefore an upper bound on the value to the bank of moving screening online, rather than forgone profit.

6.2 Increasing the Risk-Weighted Capital Requirement

Bank capital requirements are a central tool of prudential policy, and regulators continue to debate how high they should be. A recurring concern is that higher requirements may restrict

[18]The guidelines ask institutions to specify the use of automated models "in a way that is appropriate to the size, nature and complexity of the credit facility" and, "to mitigate model risk and potential errors", to have "a process for manual intervention or manual override of the automated decision" (European Banking Authority, 2020). In the consultation, industry respondents noted that the guidance "assumes a level of human intervention, which does not work for the many firms that use automated credit".

small firms' access to credit, since SMEs depend heavily on bank lending and are costly for banks to assess. The debate on the UK's implementation of the Basel 3.1 reforms illustrates this concern. The proposals included removing the SME supporting factor, a discount that lowers the capital banks must hold against SME loans by about a quarter (Appendix A), and business groups warned Parliament that the change would make finance less available and more expensive for small firms (Institute of Directors, 2023). How much would such an increase in capital requirements affect SMEs' access to credit, and which borrowers would bear it?

For comparability with existing estimates, we study a 10 percentage point increase in the bank's minimum capital ratio. This raises the capital cost of the bank's lending by 83 basis points, above the 60 basis points that Benetton (2021) finds for UK mortgages. The larger effect reflects the higher risk weights of unsecured SME loans, about 45%, compared with 27% for mortgages. This increase in cost lies within the range of 25 to 90 basis points estimated in earlier work (Hanson, Kashyap and Stein, 2011; Baker and Wurgler, 2015).

Only part of this cost increase reaches borrowers. Table 12 decomposes the change in the rates on each lender's originated loans. The bank's rates rise by 45 basis points, about 4% of the baseline rate of 10.7%. The bank absorbs much of the higher cost through a markup that falls by 31 basis points, and the borrowers it retains are slightly safer, lowering its expected default losses by 7 basis points. The non-bank faces no capital requirement, but its rates rise by 6 basis points: it raises its markup as its rival becomes more expensive, and its expected default losses rise as it takes on some of the bank's former borrowers.

These changes in rates also reallocate credit and risk. The higher requirement mainly reduces access to credit. Table 13 shows where the bank's baseline borrowers go. The bank loses 12% of its borrowers below the cap and 14% above it. Most of them stop borrowing altogether, and they are the borrowers with the highest unobserved risk. The remaining 2 to 3% move to the non-bank. Which borrowers move depends on how precisely the bank screens. Below the cap, the bank's screening is least precise. It assesses the borrowers who switch as riskier than those who stay, although they are on average safer. The higher requirement thus leads the bank to price out some borrowers it wrongly assesses as risky, and the non-bank takes them. Above the cap, the bank screens more precisely, so its assessments are closer to borrowers' true risk. There, the borrowers who switch are riskier than those who stay.

TABLE 12: Counterfactual: Increasing the Bank's Minimum Capital Ratio by 10 Percentage Points

	Bank	Non-bank
Rate on originated loans	45.0	6.0
Decomposition:		
Capital cost	83.0	0.0
Expected default loss	−7.0	2.5
Markup	−31.0	3.5
Baseline rate (%)	10.7	14.1

Note: Changes relative to the baseline equilibrium, in basis points, after a 10 percentage point increase in the bank's minimum capital ratio. Changes in the rates on originated loans reflect both changes in the rates lenders offer and changes in the composition of their borrowers. All components are effective, that is, adjusted for the probability of repayment as in equation (15). The non-bank's origination costs are unchanged, so the bank's capital cost row captures the full change in its cost of lending other than expected default losses.

TABLE 13: Counterfactual: Reallocation of the Bank's Baseline Borrowers

	Below cap			Above cap		
	Share (%)	Assessed θ	True θ	Share (%)	Assessed θ	True θ
Stay with the bank	88	−0.09	−0.11	86	−0.28	−0.20
Move to the non-bank	3	0.07	−0.33	2	0.09	−0.09
Stop borrowing	9	0.13	0.22	12	0.10	0.23

Note: Borrowers who take a bank loan in the baseline, by segment and by where they borrow after a 10 percentage point increase in the bank's minimum capital ratio. Share is the share of these borrowers in each destination. True θ is the borrower's unobserved risk, and assessed θ is the bank's posterior mean of it in the baseline; higher values indicate higher risk. Both are averaged within each group.

Taken together, a higher capital requirement has modest effects on rates but substantial effects on access to credit and on the allocation of risk. Because banks absorb part of the higher cost through lower markups, rates rise by much less than the cost of capital. Instead, the bank lends to fewer borrowers, and most of those it no longer serves stop borrowing. Since the capital charge rests on the bank's own assessment of each borrower, the burden of the requirement falls on the borrowers the bank believes are risky. Where its screening is imprecise, these include safe borrowers, who move to non-banks. How capital regulation shapes credit and risk allocation therefore depends on how precisely banks screen.[19]

[19]The counterfactual is a short-run assessment in which lenders adjust only their rates, holding fixed their screening technologies, their costs other than the capital charge, and market structure. As in Benetton (2021), our estimates can therefore be read as an upper bound on the costs of a higher requirement, since

Returning to the motivating example, removing the SME supporting factor raises the capital required against each SME loan by about 31%, equivalent to an increase of about 4 percentage points in the bank's minimum capital ratio, or roughly 40% of the shock we study. Scaling our results linearly, the removal would raise the capital cost of bank lending by about 34 basis points and bank rates by about 18, and would cause about 5% of the bank's borrowers to lose bank credit.

7 Conclusion

This paper studies competition between banks and non-banks in UK unsecured SME lending along two dimensions, bank-only risk-based capital regulation and differences in screening, and the interaction between them. Because capital requirements are risk-based, the capital a bank must hold against a loan depends on its own assessment of the borrower's risk, so screening shapes regulatory cost. Using hand-collected data on lenders' screening channels, we document that banks screen smaller loans online and larger loans through relationship managers, while non-banks screen online at every loan size, and that this switch coincides with sharp differences in how credit and risk are allocated between the two lenders. We develop and estimate a model of competition with individualised pricing to quantify these forces.

Three findings emerge. First, the ranking of screening precision depends on the screening channel: non-banks assess risk more precisely than banks when both screen online, but banks become more precise once they switch to human screening. Second, moving banks' screening online expands access to credit, but the lower precision raises banks' default rates and their regulatory cost. Third, a higher capital requirement raises banks' cost of lending. Because the capital charge rests on the bank's own assessment, the higher cost falls on borrowers it believes are risky. Where the bank screens less precisely than the non-bank, these include safe borrowers who move to the non-bank.

Taken together, our results show that in this market, capital regulation and screening are closely linked: through the risk weights, how precisely banks screen shapes how capital requirements affect rates and the allocation of credit and risk, for both banks and non-

banks could also respond through other margins, such as issuing equity, retaining earnings or adjusting deposit rates. Nor does the counterfactual capture the benefits of higher bank capital, such as greater resilience of the banking system.

banks. Our counterfactuals focus on these allocative effects as lenders adjust their rates, and complement analyses of the stability benefits of bank capital, which lie outside our framework. Extending the framework to lenders' choice of screening technology is a natural direction for future work.

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Appendix

A Regulatory Risk Weight Function

Under the Basel Internal Ratings-Based (IRB) framework, unsecured SME loans that meet specific retail criteria are classified as "Other Retail Exposures".[20] Unlike corporate exposures, the capital requirement for this asset class does not include an explicit maturity adjustment. Instead, the risk weight is determined by a continuous function of the estimated Probability of Default (PD) and Loss Given Default (LGD). The calculation proceeds in three steps. First, the regulator defines the asset correlation factor, R, which determines the sensitivity of the asset's value to systematic risk. For retail exposures, this correlation is negatively related to the probability of default, reflecting the empirical observation that higher-risk borrowers are less correlated with the general business cycle. The asset correlation R is given by:

$$R = 0.03 \left(\frac{1 - e^{-35 \cdot PD}}{1 - e^{-35}} \right) + 0.16 \left(1 - \frac{1 - e^{-35 \cdot PD}}{1 - e^{-35}} \right) \tag{16}$$

Second, the capital requirement K—representing the capital required per unit of exposure to cover unexpected losses at a 99.9% confidence level—is calculated using the Vasicek asymptotic single-risk factor model. The formula subtracts the Expected Loss ($PD \cdot LGD$) from the total conditional expected loss, ensuring that capital is held only for Unexpected Loss:

$$K = LGD \cdot N \left(\frac{G(PD) + \sqrt{R} \cdot G(0.999)}{\sqrt{1 - R}} \right) - PD \cdot LGD \tag{17}$$

where $N(\cdot)$ denotes the cumulative distribution function of the standard normal distribution, and $G(\cdot)$ denotes its inverse.

Finally, the regulation incorporates an SME Supporting Factor (SF) designed to encourage lending to small businesses. For exposures below €1.5 million, which encompasses the

[20] This includes limits on the size of the exposure to a given small business and the granularity on the exposure relative to the overall portfolio, as specified by CRE30, available at https://www.bis.org/basel_framework/standard/CRE.

unsecured retail loans in our sample, this factor reduces the capital requirement by approximately 23.81% (setting $SF = 0.7619$). The final Risk-Weighted Assets (RWA) are obtained by converting K into a risk-weight equivalent, scaling by the Exposure at Default (EAD), and applying this discount factor:

$$RWA = K \cdot 12.5 \cdot EAD \cdot SF \tag{18}$$

B Case Study: A Cap Increase Within One Bank

Design. One bank in our sample raised its online application cap from £25,000 to £50,000 in the second quarter of 2023. After the increase, loans between £25,000 and £50,000 to its customers are screened automatically rather than by a human. We use all bank and non-bank loans to firms whose main current account is with this bank, and define three loan size categories: below £25,000, which is within the cap throughout; between £25,000 and £50,000, which moves within the cap; and above £50,000, which stays above the cap throughout and serves as a placebo. $Post_t$ equals one for loans originated after the increase.

Credit allocation. We regress an indicator for a bank loan on the size categories, their interactions with $Post_t$, firm controls and fixed effects:

$$\text{Bank}_{ijt} = \sum_{s \in \{M,H\}} (\gamma_s S_{is} + \delta_s S_{is}\, Post_t) + \Gamma X_{it} + \Phi_{\text{F.E.}} + \epsilon_{ijt}, \tag{19}$$

where S_{iM} and S_{iH} indicate loans between £25,000 and £50,000 and above £50,000. The coefficient δ_M measures the change in the bank share between £25,000 and £50,000 after the increase, relative to the change for loans below £25,000, and δ_H is the placebo.

Table B.1 shows that the bank share between £25,000 and £50,000 rose by about 21 percentage points (Column 1), and by 22 points with quarter × region × industry fixed effects (Column 2). Above £50,000 the bank share did not rise; if anything it fell slightly.

Pricing. We use all loans in the sample and let T_j equal one for loans originated by the bank that raised its cap. We regress the interest rate on the size categories, $Post_t$ and T_j, together with all their interactions, the bank indicator interacted with the size categories and $Post_t$, and the same terms interacted with the default indicator, plus firm controls and fixed effects. The key coefficient is $T_j \times Post_t \times S_{iM}$. It measures how much this bank's rates on loans between £25,000 and £50,000 changed after the increase, relative to its rates on loans below £25,000, compared with the same change at all other lenders. Because most other

TABLE B.1: Case Study: Bank Share After the Cap Increase

	(1)	(2)
£25k–£50k × post, δ_M	0.209***	0.224***
	(0.068)	(0.075)
Above £50k × post, δ_H	−0.093**	−0.067
	(0.042)	(0.041)
£25k–£50k, γ_M	−0.592***	−0.604***
	(0.020)	(0.025)
Above £50k, γ_H	−0.597***	−0.620***
	(0.028)	(0.027)
Quarter, region, industry FE	✓	
Quarter × region × industry FE		✓
Maturity FE	✓	✓
Observations	14,731	12,804
R^2	0.497	0.602

Notes: OLS estimates of equation (19). The dependent variable equals one for bank loans. The sample contains all bank and non-bank loans to firms whose main current account is with the bank that raised its cap. The omitted category is loans below £25,000. Controls are quartiles of firm age and assets and deciles of credit score. Standard errors clustered at the lender × year × region level. *** p < 0.01, ** p < 0.05, * p < 0.1.

lenders are banks whose caps did not change, they face the same interest rate environment and the same pricing of loan size, but no change in screening. Table B.2 shows that after the increase, the bank's rates on loans between £25,000 and £50,000 rose by about 1.6 percentage points relative to other lenders, and by 1.8 points with lender fixed effects. Above £50,000, which remained above the cap, there is no change. The bank's premium on loans that later default fell in the same range, by a similar amount.

TABLE B.2: Case Study: Interest Rates After the Cap Increase

	(1)	(2)	(3)
$T\times$ post $\times$ £25k–£50k	1.562***	1.575***	1.794***
	(0.328)	(0.325)	(0.289)
$T\times$ post $\times$ above £50k	−0.364	−0.387	−0.242
	(0.308)	(0.305)	(0.281)
Default $\times T\times$ post $\times$ £25k–£50k		−1.624*	−1.829*
		(0.918)	(0.962)
Default interactions		✓	✓
Lender fixed effects			✓
Observations	57,548	57,548	57,547
R^2	0.578	0.580	0.590

Notes: The dependent variable is the annualised interest rate in percentage points. The sample contains all bank and non-bank loans. T equals one for loans originated by the bank that raised its cap. All lower-order terms are included and not reported. All columns include quartiles of firm age and assets, deciles of credit score, and quarter, region, sector and maturity fixed effects. Standard errors clustered at the lender × year × region level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

C Local Evidence Near the Online Application Cap

Table 3 compares lender types across the whole loan size distribution, so differences in default between the below-cap and above-cap segments could reflect differences between small and large loans rather than the cap itself. This appendix compares loans of similar size on either side of the £25,000 cap. We focus on this threshold because it offers a natural placebo group: borrowers whose bank caps at £50,000, for whom £25,000 is not a cap. If a change at £25,000 reflected loan size, it would also appear for them.

Default. Table C.1 compares loans of £15,000–£25,000 with loans of £25,000–£37,000. Predicted default is the fitted probability from the probit in Table 3, using quartiles of age and assets and deciles of credit score, and unexplained default is realised minus predicted default. For borrowers whose bank caps at £25,000, realised bank default falls from 1.8% to 0.2% across the threshold and unexplained default from −0.8 to −1.9 percentage points. Observable risk also falls above the cap, as these borrowers are larger and have higher credit scores, but by much less: predicted default falls by only 0.5 percentage points. In the placebo group, neither realised nor unexplained default falls. Non-bank default remains broadly stable, and its change across the threshold is not statistically significant. Above the cap, banks with a £25,000 cap originate 430 loans, of which one defaults. If these loans defaulted at their predicted rate, the probability of observing at most one default would be about 0.1%, and about 2.5% if they kept the unexplained default rate of the bank's loans below the cap.

TABLE C.1: Default Around £25,000 by Lender Type and Cap Group

	Loans	Realised	Predicted	Unexplained
Panel A. Loans of £15,000–£25,000				
Banks, £25k cap	3,043	1.81	2.56	−0.75
Banks, £50k cap (placebo)	2,537	1.77	2.41	−0.64
Non-banks	2,319	2.54	2.52	0.03
Panel B. Loans of £25,000–£37,000				
Banks, £25k cap	430	0.23	2.08	−1.85
Banks, £50k cap (placebo)	1,760	2.10	2.24	−0.14
Non-banks	5,929	2.92	2.48	0.44

Notes: One-year default rates in percent. Predicted default is the fitted probability from the probit of default on quartiles of age and assets, deciles of credit score, and region, sector, quarter and maturity fixed effects. Unexplained default is realised minus predicted default. Loans of exactly £25,000 are included in Panel A. Non-bank loans pool borrowers from both cap groups.

D Default Regression in the Multi-Lender Subsample

TABLE D.1: Loan Default - Lender Type

	$\mathbb{1}^{D}_{ijt} \times 100$		
	(1)	(2)	(3)
Age Category = Young	1.337	1.579	0.4869
	(1.037)	(1.044)	(1.139)
Asset Category = Below Median	1.128	1.019	1.337
	(1.023)	(1.050)	(1.459)
Credit Score Category = Normal	0.5747	0.6332	-1.666
	(0.8461)	(0.8572)	(1.276)
Credit Score Category = Safe	-0.0368	0.1879	-3.817**
	(1.161)	(1.179)	(1.753)
Bank	-1.793***	-1.736***	-2.309**
	(0.5584)	(0.5559)	(0.9036)
Observations	3,043	3,043	1,808
R^2	0.03585	0.04569	0.45191

Note: This table reports OLS estimates of the Linear Probability Model in Equation **??**, in the restricted sample of firms that borrow from multiple lenders. The dependent variable is the ex-post default indicator scaled by 100. $\mathbb{1}^{\text{Bank}}_{j}$ and $\mathbb{1}^{\text{Large}}_{it}$ are indicator variables for banks and large loans, respectively. Firm controls include categorical dummies for age, asset size, and credit score. The regression includes Region, Sector, and Quarterly Time fixed effects. The omitted baseline category defines an old firm with high assets (above median) and a risky credit score, borrowing from a non-bank. Standard errors are clustered at the Region × Sector level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

E Extended Pricing Regression with Lender-Type Interactions

To capture how the pricing of observable firm characteristics differs between banks and non-banks, we estimate an extended version of the pricing regression in Equation 1 that interacts each firm-level control with the bank indicator $\mathbb{1}_j^{\text{Bank}}$:

$$r_{ijt} = \delta_1 \mathbb{1}_j^{\text{Bank}} + \delta_2 \mathbb{1}_{ijt}^{\text{Default}} + \lambda \mathbb{1}_{ijt}^{\text{Default}} \times \mathbb{1}_j^{\text{Bank}} + \Gamma_1 X_{it} + \Gamma_2 \left(X_{it} \times \mathbb{1}_j^{\text{Bank}} \right) + \Phi_{F.E.} + \epsilon_{ijt} \,, \quad (20)$$

where X_{it} is the same vector of firm controls (above-median age, above-median assets, and credit score categories) used in the main pricing regression. The coefficients Γ_2 capture how the pricing of each firm characteristic varies between banks and non-banks. The estimates are reported in Table E.1.

The interaction coefficients reveal that banks price observable risk characteristics differently from non-banks. The negative coefficient on Bank × Young indicates that banks apply a smaller age premium than non-banks, while the positive coefficients on Bank × Credit Score (Normal) and Bank × Credit Score (Safe) indicate that banks compress the credit-score discount relative to non-banks. The asset-size interaction is small and not statistically significant. We target the full vector of coefficients from Column (5) in the structural estimation.

TABLE E.1: Extended Pricing Regression with Lender-Type Interactions

	Interest Rate (%)
Firm Controls	
Young Firm	1.198***
	(0.058)
Small Asset (Below Med.)	1.409***
	(0.062)
Credit Score (Normal)	−1.316***
	(0.068)
Credit Score (Safe)	−2.067***
	(0.074)
Lender & Outcome	
$\mathbb{1}_j^{\text{Bank}}$	−6.523***
	(0.105)
$\mathbb{1}_{ijt}^{\text{Default}}$	2.527***
	(0.191)
$\mathbb{1}_j^{\text{Bank}} \times \mathbb{1}_{ijt}^{\text{Default}}$	−0.789**
	(0.320)
Bank × Firm Controls	
$\mathbb{1}_j^{\text{Bank}}\times$ Young Firm	−0.530***
	(0.072)
$\mathbb{1}_j^{\text{Bank}}\times$ Small Asset (Below Med.)	0.022
	(0.099)
$\mathbb{1}_j^{\text{Bank}}\times$ Credit Score (Normal)	0.909***
	(0.090)
$\mathbb{1}_j^{\text{Bank}}\times$ Credit Score (Safe)	1.274***
	(0.111)
Observations	64,711
R^2	0.454

Notes: This table reports OLS estimates of the extended pricing regression in Equation 20, which interacts firm-level controls (age, asset size, and credit score) with the bank indicator $\mathbb{1}_j^{\text{Bank}}$. The dependent variable is the annualized interest rate in percentage points. $\mathbb{1}_j^{\text{Bank}}$ and $\mathbb{1}_{ijt}^{\text{Default}}$ are indicator variables for banks and default within one year of origination, respectively. The omitted category defines an old firm with above-median assets and a risky credit score, borrowing from a non-bank. The regression includes Region, Sector, Quarter, and Maturity fixed effects. Standard errors are clustered at the lender × year × region level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

F Moral Hazard

Our model assumes there is no moral hazard, that is, a direct causal effect of changes in interest rates on default. To understand the validity of this assumption, we exploit the fact that certain bank loans have floating interest rates, which increased substantially during the post-pandemic interest rate hike in the UK.

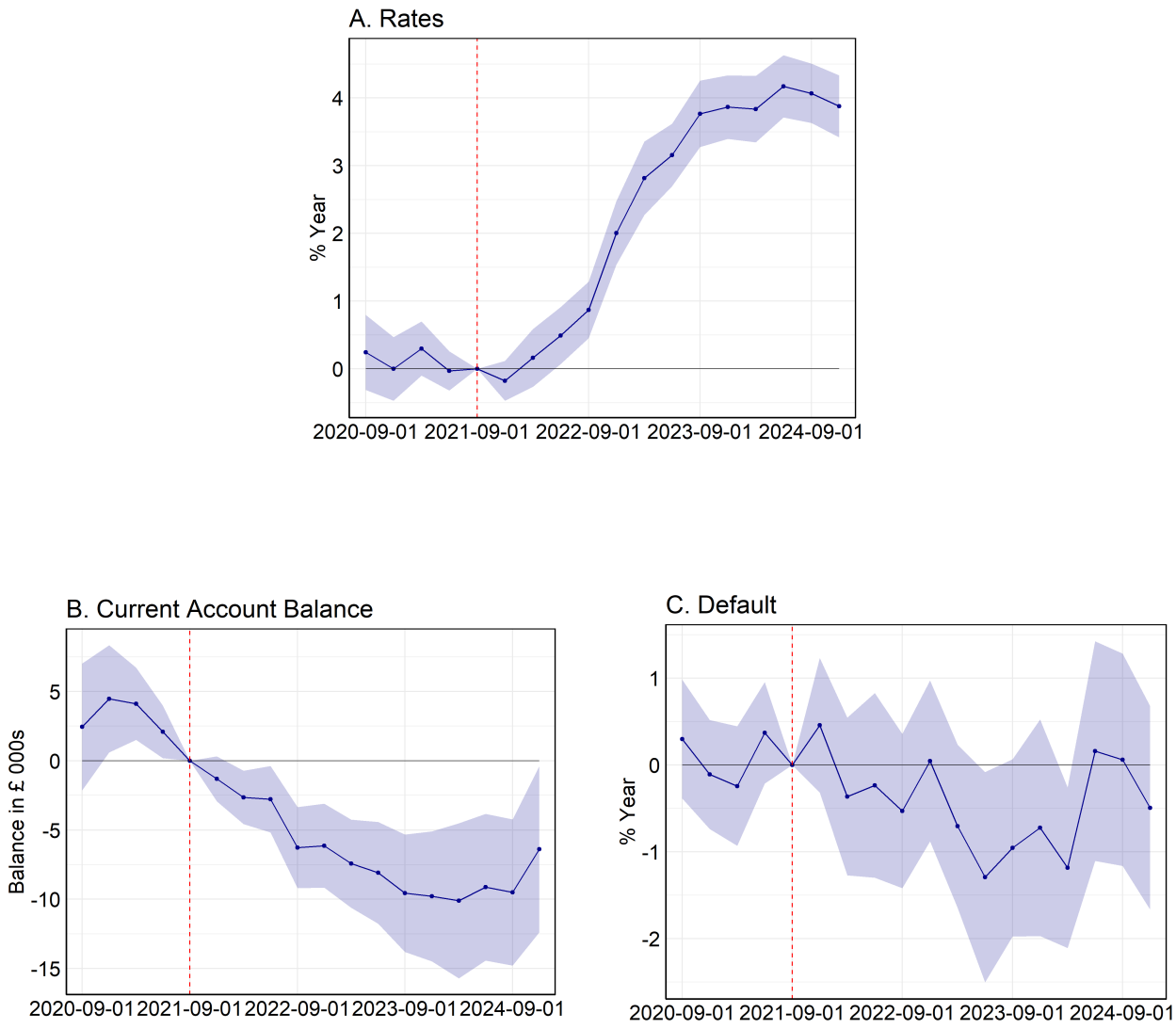
Specifically, using a firm-quarter panel, for each firm i that borrows from a bank j that issues floating rate loans, we define a dummy $Floating_i$ which equals to 1 if firm i has *any* floating rate loans issued before July 2021. We then compare outcomes of firms with and without floating rate loans using a differences-in-differences type setting:

$$y_{it} = \alpha_i + \alpha_t + \sum_{\tau \neq 0} \delta_\tau \times Floating_i + \epsilon_{it} \tag{21}$$

Where y_{it} is an outcome for firm i in quarter t, and α_i and α_t are firm and time fixed effects, respectively. The coefficients of interest δ_τ correspond to the difference between the values of y_{it} in period τ relative to the baseline period of 2021Q3, right before the start of the interest rate hike. We use three different outcomes to assess how floating rate borrowers experience differences in their cash-flows compared with fixed rate borrowers during the interest rate hike: interest rate payments, net current account balances and default. The results are shown in Figure F.1.

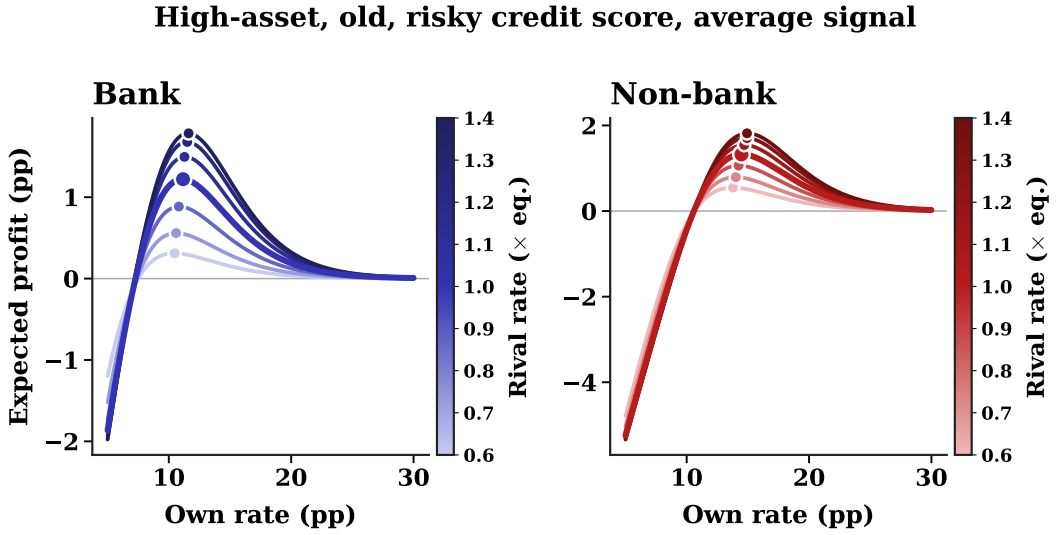
Panel A shows that firms with floating rate loans experience large increases in their interest rates compare with fixed rate borrowers, relative to the differences in September 2021, right before the interest rate hike. Such increase in interest rate expenses is associated with a decrease in net current account inflows, suggesting firms do spend more resources paying for rising monthly installments. However, despite the large increase in interest rates faced by floating rate borrowers, we see no differences in performance over time, with default rates across both groups evolving similarly after the interest rate hike. This suggests a limited role for interest rates in *causing* default.

FIGURE F.1: Floating rate versus fixed rate borrowers



Notes: This figure displays the point estimates of δ_τ from estimating specification 21, using three different outcomes: average interest rates in quarter t (Panel A); total net current account balances in quarter t in thousands of £ (Panel B), and a default dummy which equals 1 if any outstanding loan to firm i has been flagged as under default in quarter t (Panel C). Standard errors are clustered at the region-industry level. Error bands correspond to 95% confidence intervals.

FIGURE G.1: Expected Profit and the Lender's Own Rate



Note: Expected profit of each lender as a function of its own rate, holding the rival's rate at its equilibrium value, at the estimated parameters, for one borrower group and signal value.

G Properties of the Pricing Equilibrium

The equilibrium of the pricing game has no closed form, so we solve the system of lenders' first-order conditions numerically, separately for each of the 24 borrower groups for which we compute equilibrium rates. Since we cannot establish uniqueness analytically, we examine three properties of the equilibrium $z^\star$ numerically at the estimated parameters: the shape of each lender's expected profit around its equilibrium rate, the local uniqueness of the equilibrium, and the solutions obtained when the solver is restarted from other initial values.

Expected profit. Figure G.1 plots a lender's expected profit as a function of its own rate, holding the rival's rate at its equilibrium value, for one borrower group and signal value. Over the range of rates shown, expected profit is single-peaked, with its maximum at the solution of the lender's first-order condition, so the lender's problem is well behaved in this range.

TABLE G.1: Local Uniqueness of the Equilibrium

Property	Criterion	Across 24 groups	Result
Isolated equilibrium	$\sigma_{\min}(J) > 0$	[0.69, 1.15]	Nonsingular (24/24)
Local contraction	$\rho(DF) < 1$ at $z^\star$	≤ 0.15	Holds (24/24)
Contraction on box	$\sup_{\pm 40\%} \|DF\|_\infty < 1$	[0.10, 2.16]	Holds (14/24)

Note: $z^\star$ denotes the equilibrium rates. J is the Jacobian of the stacked system of first-order conditions, and $\sigma_{\min}(J)$ its smallest singular value. DF is the Jacobian of the best-response map F, $\rho(DF)$ its spectral radius, and $\sup_{\pm 40\%} \|DF\|_\infty$ the largest value of its infinity norm over rates within 40% of $z^\star$.

Local uniqueness. Table G.1 reports three properties of the equilibrium. First, the Jacobian J of the stacked system of first-order conditions is nonsingular at $z^\star$ in all 24 groups, so the equilibrium is an isolated solution of the system. Second, the best-response map F, which solves each lender's first-order condition for its own rate given the rival's, is a local contraction at $z^\star$ in all groups, since the spectral radius of its Jacobian DF is below one; the equilibrium is therefore locally unique. Third, in 14 of the 24 groups, F is a contraction over the box of rates within 40% of $z^\star$, so in these groups the equilibrium is the only fixed point within that box.

Restarts of the solver. Table G.2 reports the results of restarting the solver from other initial values. We use two sets of starting points: perturbations of the equilibrium by between 2% and 50%, and rates drawn uniformly between 5 and 30 percentage points. Across 1,440 restarts, the solver converges in every case, and no restart yields a solution that differs from $z^\star$ by more than 10^{-2}.

These checks show that the equilibrium we compute is locally unique and that the solver returns it from a wide range of starting values. They do not rule out other equilibria outside the range we examine. Following Cosconati et al. (2025), we therefore solve each counterfactual starting from the estimated equilibrium.

TABLE G.2: Equilibrium Across Restarts of the Solver

	Perturbed starts	Random starts
Initial values	$\pm 2\%$ to $\pm 50\%$ of $z^\star$	Uniform on $[5, 30]$ pp
Restarts across 24 groups	720	720
Converged to a solution	100%	100%
Maximum $\|z_{\text{restart}} - z^\star\|$ (pp)	1.8×10^{-9}	4.2×10^{-9}
Total restarts	1,440	
Distinct solutions found	0	

Note: From each starting point, the system of first-order conditions is solved again. A solution counts as distinct if it differs from $z^\star$ by more than 10^{-2}.

TABLE H.1: Remaining Parameter Estimates

		Estimate	Standard error	Implied loan size coeff
Demand				
Non-price term, bank	δ_B^D	3.919	0.192	
Non-price term, non-bank	δ_N^D	4.699	0.349	
Age: young	β^D	0.437	0.097	
Assets: below median		0.065	0.218	
Credit score: normal		−0.078	0.058	
Credit score: safe		−0.638	0.129	
Default				
Unobserved risk	β_θ^F	3.290	0.882	
Non-bank default shifter	δ_N^F	−8.307	2.193	
Bank shifter relative to non-bank	$\delta_B^F - \delta_N^F$	2.440	0.999	
Age: young	β^F	1.194	0.258	
Assets: below median		0.628	0.238	
Credit score: normal		−0.377	0.211	
Credit score: safe		−2.371	0.703	
Public signal from loan size				
Constant		−13.651	11.160	2.205
Age: young		4.708	3.473	−0.761
Assets: below median		6.168	5.634	−0.997
Credit score: normal		−2.035	1.676	0.329
Credit score: safe		−2.933	2.413	0.474
Log loan size	$1/\beta_\theta^\ell$	6.190	5.235	$\beta_\theta^\ell = 0.162$
Signal noise s.d.	$\sigma_\ell/\beta_\theta^\ell$	8.965	7.041	$\sigma_\ell = 1.448$

Note: The public signal is estimated in the form $\tilde{s}_i = b_0 + b'X_i + b_\ell \log \ell_i + \tilde{\epsilon}_i$, where $b_\ell = 1/\beta_\theta^\ell$ and $b = -\beta^\ell/\beta_\theta^\ell$. The last column reports the implied coefficients of the loan size equation (6). The bank's default shifter is $\delta_N^F + (\delta_B^F - \delta_N^F)$.

H Remaining Parameter Estimates

I Model Fit: All Targeted Moments

Table I.1 reports all targeted moments in the data and in the model. The moments are grouped by the auxiliary statistic they come from: the pricing regressions, the default regression, the loan size regression, default rates by lender type and segment, and market shares. The early and later periods refer to the parts of our sample before and after one of the large non-banks stopped raising funds from retail investors, which persistently widened the gap between bank and non-bank rates. Lenders' costs are allowed to differ across the two periods, and the market shares and the bank-non-bank rate gap in each period help identify borrowers' price sensitivity (Section 5.2).

TABLE I.1: Model Fit: All Targeted Moments

	Data	Model
Pricing regression by lender type and segment		
Constant	12.093	11.535
Bank, below cap	−3.425	−3.823
Bank, above cap	−6.091	−6.854
Non-bank, above cap	0.877	1.140
Age: young	0.273	1.202
Credit score: normal	0.700	1.173
Credit score: risky	0.974	1.627
Assets: below median	0.300	0.554
Default × bank, below cap	1.026	0.780
Default × bank, above cap	1.395	1.857
Default × non-bank, below cap	2.051	1.488
Default × non-bank, above cap	1.057	1.348
Bank × early period	2.669	2.378

	Data	Model
Pricing regression with bank interactions		
Bank × age: young	0.278	0.592
Age: young	0.227	1.009
Bank × assets: below median	1.906	1.017
Assets: below median	−0.053	0.262
Bank × credit score: normal	−0.223	−0.357
Bank × credit score: safe	0.095	−0.070
Credit score: normal	−0.198	−0.298
Credit score: safe	−1.041	−1.550
Bank	−5.498	−5.279
Bank × default	−0.187	−0.102
Default	1.280	1.212
Default regression		
Constant	2.841	2.364
Bank	−1.358	−1.723
Age: young	0.735	0.963
Assets: below median	0.313	0.477
Credit score: normal	−0.811	−0.382
Credit score: safe	−1.247	−1.581
Loan size regression		
Constant	4.242	4.724
Default	0.097	0.335
Age: young	−0.032	−0.889
Assets: below median	−1.097	−1.140
Credit score: normal	0.141	0.378

	Data	Model
Credit score: safe	0.345	0.607
Default rates (%)		
Non-bank, below cap	2.568	2.900
Non-bank, above cap	2.812	2.468
Bank, below cap	1.869	1.767
Bank, above cap	0.322	0.607
Shares		
Non-bank, age old, later period	0.416	0.311
Bank, age old, later period	0.253	0.187
No loan, age old, later period	0.332	0.502
Non-bank, age young, later period	0.230	0.176
Bank, age young, later period	0.230	0.339
No loan, age young, later period	0.540	0.485
Non-bank, age old, early period	0.536	0.626
Bank, age old, early period	0.141	0.139
No loan, age old, early period	0.323	0.235
Non-bank, age young, early period	0.306	0.389
Bank, age young, early period	0.147	0.277
No loan, age young, early period	0.547	0.334
Non-bank, assets above median, later period	0.529	0.311
Bank, assets above median, later period	0.262	0.184
No loan, assets above median, later period	0.209	0.505
Non-bank, assets below median, later period	0.189	0.179
Bank, assets below median, later period	0.227	0.338
No loan, assets below median, later period	0.584	0.484

	Data	Model
Non-bank, assets above median, early period	0.659	0.617
Bank, assets above median, early period	0.175	0.137
No loan, assets above median, early period	0.167	0.246
Non-bank, assets below median, early period	0.216	0.399
Bank, assets below median, early period	0.125	0.275
No loan, assets below median, early period	0.658	0.326
Non-bank, credit score normal, later period	0.417	0.266
Bank, credit score normal, later period	0.246	0.243
No loan, credit score normal, later period	0.337	0.491
Non-bank, credit score risky, later period	0.200	0.167
Bank, credit score risky, later period	0.221	0.344
No loan, credit score risky, later period	0.579	0.489
Non-bank, credit score safe, later period	0.420	0.344
Bank, credit score safe, later period	0.294	0.155
No loan, credit score safe, later period	0.286	0.501
Non-bank, credit score normal, early period	0.532	0.542
Bank, credit score normal, early period	0.143	0.187
No loan, credit score normal, early period	0.325	0.272
Non-bank, credit score risky, early period	0.254	0.377
Bank, credit score risky, early period	0.131	0.283
No loan, credit score risky, early period	0.615	0.341
Non-bank, credit score safe, early period	0.559	0.687
Bank, credit score safe, early period	0.199	0.111
No loan, credit score safe, early period	0.243	0.202
Non-bank share, below cap	0.267	0.277

	Data	Model
Non-bank share, above cap	0.809	0.773
Share of loans above cap	0.614	0.488